International Journal of Solids and Structures 339 (2026) 114156

Contents lists available at ScienceDirect

International Journal of Solids and Structures

journal homepage: www.elsevier.com/locate/ijsolstr

A thermodynamically consistent two-scale thermo–elastic–damage model derived via asymptotic homogenization

Quanzhang Li [a,1], Yipeng Rao [b,1], Junzhi Cui [c], Meizhen Xiang [a,*]

[a] National key Laboratory of Computational Physics, Institute of Applied Physics and Computational Mathematics, Beijing, 100088, China
[b] Key Laboratory of Impact and Safety Engineer, Ministry of Education, Ningbo University, Ningbo, 315211, China
[c] LSEC, ICMSEC, Academy of Mathematics and Systems Science, CAS, Beijing, 100090, China

ARTICLE INFO

Keywords:
Asymptotic homogenization
Thermo–mechanical brittle fracture
Heat dissipation
Thermodynamic consistency

ABSTRACT

This work develops a thermodynamically consistent thermo–elastic–damage model within a two-scale asymptotic homogenization framework, bridging microcrack dynamics and macroscopic fracture behavior. Starting from a representative volume element containing an embedded microcrack, the homogenized Helmholtz free and kinetic energy densities are rigorously derived through two-scale asymptotic expansion. Application of the first and second laws of thermodynamics yields a set of nonlocal thermo–elastic coupling equations, while incorporating Griffith's criterion leads to a dynamic damage evolution law linking microcrack growth with the macroscopic thermomechanical state (strain, strain gradient, strain rate, and temperature). During crack propagation, the excess internal energy beyond that required for new surface formation is dissipated as heat. The effective material properties are expressed as nonlinear functions of microcrack length and orientation via first-order cell functions, capturing anisotropic stiffness and conductivity degradation. The framework strictly satisfies the dissipation inequality, ensuring thermodynamic consistency. Numerical simulations successfully reproduce key experimental observations, including localized temperature rise near the crack front, anisotropic degradation of thermal conductivity, and strain-rate-dependent fracture responses. The predicted temperature evolution shows good quantitative agreement with reported experimental measurements.

1. Introduction

Fracture behavior is inherently a multi-scale phenomenon. Macroscale fracture is influenced by microstructural features such as voids, defects, dislocations, grains, and grain boundaries (Freund, 1990; Grassl et al., 2012; Shen et al., 2025; Ramalingam et al., 2025). Among these features, microcracks play a central role in the fracture of brittle materials such as concrete, rock, and ceramics (Nirmalendran and Horii, 1992; Moore and Lockner, 1995; Challamel, 2010). It is widely recognized that the initiation, propagation, and branching of macrocracks are largely governed by the nucleation, growth, and interaction of microcracks (Ravi-Chandar and Yang, 1997).

Accurately predicting brittle fracture requires models that incorporate microstructural effects. By embedding micromechanical descriptions of microcracks into the classical framework of fracture mechanics (Griffith, 1921; Irwin, 1958; Rice, 1968), many complex fracture phenomena can be systematically rationalized, such as the development of multiple crack branches (Ravi-Chandar and Knauss, 1984a,b,c; Knauss and Ravi-Chandar, 1985). This recognition has motivated extensive researches on brittle fracture that explicitly accounting for microcrack effects. For example, Ashby and Hallam (1986) proposed a damage mechanics framework for brittle fracture under compression, based on a critical stress criterion that explicitly incorporates crack length, orientation, and friction. Gambarotta and Lagomarsino (1993) introduced a model that accounts for unilateral frictional contact between crack faces, thereby distinguishing brittle and quasi-brittle responses under tension and compression, while emphasizing anisotropy and loading-path dependence. Paliwal and Ramesh (2008) developed a model that explicitly accounts for microcrack interactions under compressive loading by incorporating a pre-existing flaw distribution. In their framework, macroscopic inelastic deformation is attributed to the nucleation and growth of tensile "wing" microcracks. Recently, Xiang et al. (2025) developed a two-scale microcrack band damage model that integrates the phase-field approach with micromechanical damage theory. In this framework, the macroscale crack surface is explicitly defined through an equivalent transformation of the microcrack distribution, rather than through the classical regularization of a sharp crack topology. A broader overview of microcrack-based modeling

* Corresponding author.
E-mail address: xiang_meizhen@iapcm.ac.cn (M. Xiang).
[1] Contribute equally to this work.

https://doi.org/10.1016/j.ijsolstr.2026.114156
Received 16 December 2025; Received in revised form 26 March 2026; Accepted 5 June 2026
Available online 12 June 2026

Symbol	Definition
ε	Microstructural period (microcrack spacing).
l, θ	Microcrack length and inclination angle.
$d = l/\varepsilon$	Normalized microcrack length (damage parameter).
Y^{ε}, Y	Reference cell and normalized unit cell.
C^{ε}, C	Microcrack in Y^{ε} and its normalized counterpart.
$\boldsymbol{x}$, $\boldsymbol{y} = \boldsymbol{x}/\varepsilon$	Macro- and micro-spatial coordinates.
$\boldsymbol{u}^{\varepsilon}$	Displacement field.
$\mathcal{E}_{ij}$, $\mathcal{E}_{ijk}$	Macroscopic strain and strain-gradient tensors.
Σ_{ij}, Γ_{ijk}	Cauchy stress and higher-order (double) stress.
T^{ε}, T_0	Temperature field and reference temperature.
η, $\mathcal{D}$	Macroscopic entropy density and dissipation density.
E, ν, ρ	Young's modulus, Poisson's ratio, mass density.
c, α, κ	Specific heat, thermal expansion coefficient, conductivity tensor.
G_c, C_R	Critical fracture energy and crack speed parameter.
$N_i^{pq}(\boldsymbol{y})$, $P_i(\boldsymbol{y})$, $M_i(\boldsymbol{y})$	First-order cell functions.
$\hat{C}_{ijkl}$	Effective elastic stiffness tensor.
$\hat{D}_{ijklmn}$	Strain-gradient stiffness tensor.
$\hat{F}_{ijkl}$	Micro-inertial tensor.
$\hat{\kappa}_{ij}$	Effective thermal conductivity tensor.
$\hat{\beta}_{ij}$, $\hat{G}$	Thermo–mechanical coupling tensor and thermal modulus.
$w_{\varepsilon}^{\mathrm{in}}$, $w_{\varepsilon}^{\mathrm{f}}$, $w_{\varepsilon}^{\mathrm{k}}$	Microscopic internal, Helmholtz free, and kinetic energies.
w^{in}, w^{f}, w^{k}	Macroscopic internal, Helmholtz free, and kinetic energies.
RVE	Representative volume element.

approaches and their applications to brittle and quasi-brittle fracture can be found in Kranz (1983), Nemat-Nasser and Obata (1988), Meguid et al. (1991), Grah et al. (1996), Stolkarts et al. (1999), Gross and Seelig (2006), Biswas et al. (2016), Qingliang et al. (2023), Chen et al. (2024).

When the largest microstructural scale in a material approaches the relevant macroscopic length scale, classical continuum mechanics — which assumes a local relationship between stress and strain — fails to account for significant scale effects arising from nonlocal interactions at the microscale (Peerlings et al., 1996). Incorporating strain-gradient terms in constitutive equations has been confirmed to be an efficient way to capture nonlocal effects (Aifantis, 1999; Lam et al., 2003). In addition, under dynamic loading, strain rate is a crucial factor that influences fracture behaviors (Ožbolt et al., 2014; Keita et al., 2014; Erzar and Buzaud, 2012; Zinszner et al., 2015; Ivančević and Ratković, 2025; Shang and Yang, 2025). Extensive efforts have been devoted to developing mechanical models that account for strain-gradient effects (Chen et al., 1999; Zhang et al., 2020; Yang et al., 2022; Lazar, 2024) and strain-rate effects (Kipp et al., 1980; Lambert and Ross, 2000; Verleysen and Peirs, 2017; Delfani and Bidi, 2024; Chen et al., 2025) on fracture. For instance, Li (2011), Li et al. (2011) derived a strain-gradient-enhanced constitutive relation for two-dimensional elastic solids containing distributed microcracks, from which they formulated a damage evolution law explicitly incorporating strain-gradient effects. In addition, Placidi and Barchiesi (2018) proposed a variational-inequality formulation for quasi-static brittle crack propagation in isotropic two-dimensional geometrically nonlinear continua, where the surface energy density depends on the strain gradient rather than on the gradient or Laplacian of the damage field. On the strain rate side, Lambert and Ross (2000) developed a experimental procedure and accompanying theoretical analysis to characterize dynamic fracture properties of quasi-brittle materials, showing that both the effective fracture toughness and specimen strength increase markedly with loading rate. More recently, Pandya et al. (2020) proposed an isotropic neural-network-based fracture initiation model, trained and validated to capture the onset of fracture across a wide range of stress states, strain rates, and temperatures.

Additional complexity in fracture mechanics arises from thermal effects. On the one hand, temperature influences both the fracture strength (Funatsu et al., 2004; Banea et al., 2012; Feng et al., 2017) and the crack propagation path (Tangella et al., 2022; Ruan et al., 2023). On the other hand, numerous experiments have shown that brittle fracture is often accompanied by a pronounced temperature rise in a highly localized region surrounding the crack tip (Weichert and Schönert, 1974; Fuller et al., 1975; Rittel, 2000; Bjerke et al., 2002; Pandey and Chand, 2008). For instance, in Zr-based metallic glasses, a temperature increase of up to 600 °C has been observed within less than 100 μs during crack propagation (Das et al., 2018). To describe such localized heating phenomenon, several extensions of the heat conduction equation have been proposed. Mason et al. (1994) demonstrated that a fraction of the plastic work is dissipated as heat, while Sun and Hsu (1996) incorporated thermoelastic, thermoplastic, and thermofracture coupling terms by introducing a Δ-type heat source representing the difference between released energy and fracture surface energy. Cohesive-zone models with thermal dissipation have also been developed, demonstrating excellent predictive capability for dynamic fracture in PMMA (Bjerke and Lambros, 2003) and later extended to ductile fracture (Fagerström and Larsson, 2008). More recently, the phase-field approach has emerged as a versatile framework for fracture with thermal effects (Kuhn and Müller, 2009; Miehe et al., 2015b,a; Ruan et al., 2023).

To rigorously incorporate microstructural heterogeneities and coupled thermo–mechanical interactions in periodically heterogeneous continua, the two-scale asymptotic homogenization method—also referred to as mathematical homogenization theory (Bensoussan et al., 1979; Bakhvalov and Panasenko, 1989; Oleïnik et al., 1992)—provides a systematic multiscale framework (Bacigalupo et al., 2022; Dong et al., 2023; Dehghani et al., 2024). In the context of fracture mechanics, this framework has been progressively extended to enable a rigorous derivation of homogenized energy functionals, effective constitutive relations, and macroscopic crack-driving forces. Early contributions (Dascalu et al., 2008; Keita et al., 2014) established two-scale asymptotic fracture formulations based on the J-integral, thereby capturing intrinsic size effects such as strain localization, strength variability, and dispersive wave phenomena. Subsequent developments incorporated thermo–mechanical coupling, accounting for the partial conversion of fracture dissipation at propagating crack tips into heat (Dascalu and Gbetchi, 2019). More recently, asymptotic homogenization has been combined with phase-field fracture models to characterize periodically heterogeneous materials (Fantoni et al., 2020; Ma et al., 2024; Liu et al., 2024).

This approach enables the derivation of anisotropic effective elastic and fracture properties and clarifies the role of microscale damage mechanisms in governing macroscopic fracture behavior. Furthermore, You et al. (2025) integrated the variational asymptotic homogenization method — whose theoretical foundations have been substantially advanced in recent years (Smyshlyaev and Cherednichenko, 2000; Zhong et al., 2015; Srivastava et al., 2023; Pitchai et al., 2023) — with a phase-field fracture framework, thereby systematically incorporating higher-order strain-gradient effects into the fracture evolution process.

In our previous work, purely mechanical brittle fracture in microcracked materials was rigorously incorporated into the two-scale asymptotic homogenization framework. By employing homogenized strain and kinetic energies as energetic bridges across scales, microstructural effects, strain gradients, and strain-rate dependence were systematically embedded into macroscopic fracture laws and the homogenized elastodynamic equations (Rao et al., 2022, 2023; Li et al., 2025b). Nevertheless, fracture processes are intrinsically coupled with thermal effects: crack propagation generates heat dissipation, leading to local temperature rises, while temperature variations in turn influence crack initiation and the material's fracture strength.

Building on this foundation, the present study formulates a fully coupled, thermodynamically consistent, nonlocal thermo–mechanical brittle fracture model within the two-scale asymptotic homogenization framework, explicitly incorporating microcrack-induced heat dissipation and its influence on macroscopic fracture behavior. Starting from a reference cell containing embedded microcracks, the macroscopic free and kinetic energy densities are derived in a mathematically rigorous manner. This two-scale formulation naturally incorporates microcrack-induced nonlocality, capturing the effects of the intrinsic microstructural length scale, strain gradients, strain rates, and temperature, all without the need for any supplementary phenomenological assumptions. Combined with the Griffith fracture criterion, this energy formulation leads to a microcrack evolution law governed by macroscopic strain gradients, strain rates, and temperature rise. By enforcing the first and second laws of thermodynamics, the framework yields nonlocal thermo–mechanical coupling equations, in which the fracture energy dissipated at moving crack tips — defined as the internal energy exceeding that required for the formation of new microcrack surfaces — is converted into heat. The effective material parameters emerge as nonlinear functions of microcrack length and orientation through first-order cell functions, thereby capturing the anisotropic degradation of material properties.

The model is formulated entirely in terms of consistent macroscopic free and kinetic energy densities and is shown to satisfy the energy dissipation inequality derived from the first and second laws of thermodynamics, ensuring thermodynamic consistency. Its predictive capabilities are demonstrated through numerical simulations of local responses, crack propagation, and temperature fields in macroscopic brittle fracture. In particular, the predicted temperature variations at selected points along the crack path exhibit good agreement with experimental measurements.

The remainder of this paper is organized as follows. Section 2 presents the formulation of the two-scale analysis together with the macroscopic free energy and kinetic energy densities. Section 3 derives the microcrack evolution law and the coupled thermo–mechanical governing equations, and verifies the thermodynamic consistency of the proposed model. Section 4 provides numerical validations and comparisons with experimental results.

2. Two-scale analysis

2.1. Asymptotic expansions

Consider a solid domain $\Omega \subset \mathbb{R}^2$ containing a high-density population of microcracks. In realistic materials, the distribution, size, and orientation of microcracks are inherently stochastic. In the present framework, microcracks are assumed to be locally periodic: each macroscopic material point is associated with a micro-region where microcracks are idealized as straight, periodically arranged defects. The local microcrack configuration is described by the triplet (ε, l, θ), where ε is the spacing between adjacent microcracks, l their length, and θ the inclination angle relative to the global x-axis (see Fig. 1). This commonly adopted assumption in micromechanics (François and Dascalu, 2010; Wrzesniak et al., 2015) captures the essential influence of microcrack configuration while preserving both analytical tractability and computational efficiency.

Within this framework, crack propagation is modeled as a self-similar growth process: the crack orientation θ and the mean spacing ε remain fixed, while the crack length l evolves during loading. To quantify the associated microscopic damage, we introduce the normalized crack length

$$d = \frac{l}{\varepsilon} \in [0, 1]. \tag{1}$$

The limiting case $d = 1$ corresponds to a fully developed microcrack spanning the entire reference cell, signifying the onset of macroscopic material failure.

In the context of two-scale asymptotic modeling, the macroscopic mechanical response at a point $\boldsymbol{x}$ is represented by a representative volume element (RVE), or reference cell, of size ε, denoted Y_ε, as shown in Fig. 1. The microcrack embedded within the RVE is denoted by C_ε. Accordingly, the intact (solid) portion of the reference cell is defined as $Y_\varepsilon^s = Y_\varepsilon \setminus C_\varepsilon$. This RVE construction encapsulates the essential microstructural features.

Within the solid region Y_ε^s, the displacement field $\boldsymbol{u}^\varepsilon$ and the temperature field T^ε satisfy the coupled thermoelastic system composed of the linear momentum balance and the heat conduction equation:

$$\begin{cases} \rho \dfrac{\partial^2 u_i^\varepsilon}{\partial t^2} - \dfrac{\partial \sigma_{ij}^\varepsilon}{\partial x_j} = 0, & \text{(2a)} \\ c \dfrac{\partial T^\varepsilon}{\partial t} + \beta T^\varepsilon \dfrac{\partial e_{jj}(\boldsymbol{u}^\varepsilon)}{\partial t} + \dfrac{\partial q_j^\varepsilon}{\partial x_j} = 0, & \text{(2b)} \end{cases}$$

where ρ is the mass density, and the infinitesimal strain tensor is defined as

$$e_{ij}(\boldsymbol{u}^\varepsilon) = \frac{1}{2}\left(\frac{\partial u_i^\varepsilon}{\partial x_j} + \frac{\partial u_j^\varepsilon}{\partial x_i}\right).$$

The stress tensor σ_{ij}^ε is related to the strain field through the constitutive relation

$$\sigma_{ij}^\varepsilon = C_{ijkl}\, e_{kl}(\boldsymbol{u}^\varepsilon) - \beta\, \delta_{ij}\, (T^\varepsilon - T^0),$$

where C_{ijkl} is the fourth-order elasticity tensor, T^0 denotes the reference temperature, δ_{ij} is the Kronecker delta, and β quantifies the thermoelastic coupling coefficient. The heat flux $\boldsymbol{q}^\varepsilon$ is governed by Fourier's law:

$$q_j^\varepsilon = -\kappa \frac{\partial T^\varepsilon}{\partial x_j},$$

with κ denoting the thermal conductivity.

In the present formulation, microcrack propagation is considered without accounting for crack closure or frictional dissipation. Moreover, the crack surfaces are assumed to be adiabatic. Accordingly, the boundary conditions on the crack surface C_ε are prescribed as

$$\boldsymbol{\sigma}^\varepsilon \cdot \boldsymbol{n} = \boldsymbol{0}, \quad [\boldsymbol{u}^\varepsilon \cdot \boldsymbol{n}] > 0, \quad \boldsymbol{q}^\varepsilon \cdot \boldsymbol{n} = 0, \tag{3}$$

where $\boldsymbol{n}$ denotes the unit outward normal to the crack surface, and $[\cdot]$ represents the jump across the crack interface.

To facilitate the two-scale analysis, we introduce the microscopic variable $\boldsymbol{y} = \boldsymbol{x}/\varepsilon$, which maps the physical reference cell onto the normalized unit cell $Y = [0,1] \times [0,1]$, as illustrated in Fig. 2. Under this transformation, the microcrack is denoted by $C \subset Y$, with its normalized length defined as $d = \frac{l}{\varepsilon} \in [0, 1]$. The solid portion of the unit cell is then given by $Y^s = Y \backslash C$.

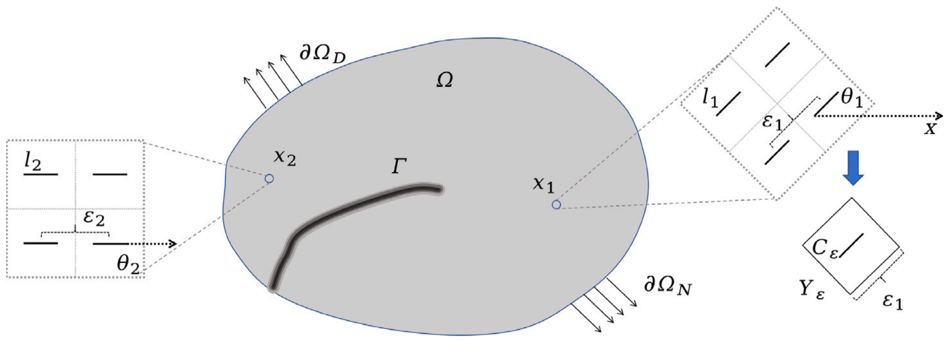


Fig. 1. Illustration of the two-scale modeling framework.

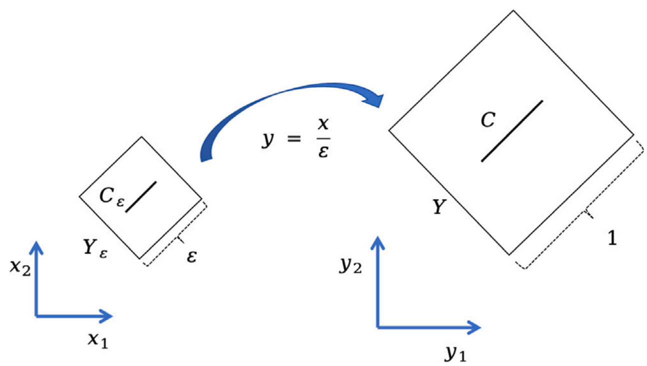


Fig. 2. Scale transformation.

For any function $f(\boldsymbol{x}, \boldsymbol{y})$ defined on the two-scale domain, the corresponding ε-dependent function is expressed as $f^{\varepsilon}(\boldsymbol{x}) = f\left(\boldsymbol{x}, \frac{\boldsymbol{x}}{\varepsilon}\right)$. Applying the chain rule, the gradient of f^{ε} with respect to the macroscopic variable $\boldsymbol{x}$ reads

$$\nabla_{\boldsymbol{x}} f^{\varepsilon}(\boldsymbol{x}) = \frac{\partial f}{\partial \boldsymbol{x}} + \frac{1}{\varepsilon}\frac{\partial f}{\partial \boldsymbol{y}}. \tag{4}$$

Accordingly, we define the infinitesimal strain tensors with respect to the macroscopic and microscopic coordinates as

$$e_{xij}(f) = \frac{1}{2}\left(\frac{\partial f_i}{\partial x_j} + \frac{\partial f_j}{\partial x_i}\right), \quad e_{yij}(f) = \frac{1}{2}\left(\frac{\partial f_i}{\partial y_j} + \frac{\partial f_j}{\partial y_i}\right). \tag{5}$$

For the subsequent two-scale analysis, we define the Sobolev space

$$\mathcal{D}(Y^s) = \left\{ \boldsymbol{u} \in [H^1(Y^s)]^2 \,\middle|\, \boldsymbol{u} \text{ is } Y\text{-periodic in } \boldsymbol{y}, \quad \int_{Y^s} \boldsymbol{u}\, \mathrm{d}\boldsymbol{y} = \boldsymbol{0} \right\}.$$

Following the standard method of asymptotic expansions (Cioranescu, 1999; Yang et al., 2015), the displacement and temperature fields are formally expanded as

$$\begin{cases} \boldsymbol{u}^{\varepsilon}(\boldsymbol{x}, t) = \boldsymbol{u}^{(0)}\left(\boldsymbol{x}, \dfrac{\boldsymbol{x}}{\varepsilon}, t\right) + \varepsilon \boldsymbol{u}^{(1)}\left(\boldsymbol{x}, \dfrac{\boldsymbol{x}}{\varepsilon}, t\right) + \varepsilon^2 \boldsymbol{u}^{(2)}\left(\boldsymbol{x}, \dfrac{\boldsymbol{x}}{\varepsilon}, t\right) + \cdots, & \text{(6a)} \\ T^{\varepsilon}(\boldsymbol{x}, t) = T^{(0)}\left(\boldsymbol{x}, \dfrac{\boldsymbol{x}}{\varepsilon}, t\right) + \varepsilon T^{(1)}\left(\boldsymbol{x}, \dfrac{\boldsymbol{x}}{\varepsilon}, t\right) + \varepsilon^2 T^{(2)}\left(\boldsymbol{x}, \dfrac{\boldsymbol{x}}{\varepsilon}, t\right) + \cdots, & \text{(6b)} \end{cases}$$

where the functions $\boldsymbol{u}^{(i)}(\boldsymbol{x}, \boldsymbol{y}, t)$ and $T^{(i)}(\boldsymbol{x}, \boldsymbol{y}, t)$, $i \in \mathbb{N}$, are Y-periodic in the microscopic variable $\boldsymbol{y}$.

By substituting the asymptotic expansions (6a), (6b) into the governing Eqs. (2a), (2b), together with the boundary condition (3), and equating the coefficients of identical powers of the small parameter ε, the leading-order terms of order ε^{-2} imply that both the zeroth-order displacement field $\boldsymbol{u}^{(0)}$ and the zeroth-order temperature field $T^{(0)}$ are independent of the microscopic variable $\boldsymbol{y}$ (Cioranescu, 1999; Li et al., 2025a).

At the subsequent order ε^{-1}, the resulting cell problems indicate that the first-order correctors $\boldsymbol{u}^{(1)}$ and $T^{(1)}$ admit a separated-variable representation of the form

$$\begin{cases} \boldsymbol{u}^{(1)}(\boldsymbol{x}, \boldsymbol{y}, t) = \boldsymbol{N}^{pq}(\boldsymbol{y})\, e_{xpq}\left(\boldsymbol{u}^{(0)}\right)(\boldsymbol{x}, t) + \boldsymbol{P}(\boldsymbol{y})\left(T^{(0)}(\boldsymbol{x}, t) - T^0\right), & \text{(7a)} \\ T^{(1)}(\boldsymbol{x}, \boldsymbol{y}, t) = M_i(\boldsymbol{y})\, \dfrac{\partial T^{(0)}}{\partial x_i}(\boldsymbol{x}, t), & \text{(7b)} \end{cases}$$

where the superposed dot $\dot{g} = \partial g / \partial t$ denotes the time derivative of a generic function g. The first-order cell functions $\boldsymbol{N}^{pq}(\boldsymbol{y})$, $\boldsymbol{P}(\boldsymbol{y})$, and $M_i(\boldsymbol{y})$, with indices $p, q, i = 1, 2$, belong to the space $\mathcal{D}(Y^s)$ and are determined by the associated microscopic cell problems detailed below.

$$\begin{cases} \dfrac{\partial}{\partial y_j}\left(C_{ijkl}\, e_{ykl}(\boldsymbol{N}^{pq}) + C_{ijpq}\right) = 0, & \text{in } Y^s, \\ \left(C_{ijkl}\, e_{ykl}(\boldsymbol{N}^{pq}) + C_{ijpq}\right) n_j = 0, & \text{on } C^{\pm}, \end{cases} \tag{8a}$$

$$\begin{cases} \dfrac{\partial}{\partial y_j}\left(C_{ijkl}\, e_{ykl}(\boldsymbol{P}) + \beta\, \delta_{ij}\right) = 0, & \text{in } Y^s, \\ \left(C_{ijkl}\, e_{ykl}(\boldsymbol{P}) + \beta\, \delta_{ij}\right) n_j = 0, & \text{on } C^{\pm}, \end{cases} \tag{8b}$$

$$\begin{cases} \dfrac{\partial}{\partial y_j}\left(\kappa\, \dfrac{\partial M_i}{\partial y_j} + \kappa\, \delta_{ij}\right) = 0, & \text{in } Y^s, \\ \left(\kappa\, \delta_{ij} + \kappa\, \dfrac{\partial M_i}{\partial y_j}\right) n_j = 0, & \text{on } C^{\pm}. \end{cases} \tag{8c}$$

2.2. Macroscopic energy densities

In this section, we derive the macroscopic energy densities at macroscopic point $\boldsymbol{x}$, including the Helmholtz free energy, and kinetic energy, based on the energy integral over the associated unit cell Y^s.

We start with the homogenized Helmholtz free energy. The microscopic Helmholtz free energy density in Y^s reads

$$w_{\varepsilon}^{\mathrm{f}}(\boldsymbol{x}, \boldsymbol{y}) = \frac{1}{2} C_{ijkl}\, e_{ij}(\boldsymbol{u}^{\varepsilon})\, e_{kl}(\boldsymbol{u}^{\varepsilon}) - \beta\, \delta_{ij}\, e_{ij}(\boldsymbol{u}^{\varepsilon})(T^{\varepsilon} - T^0) + c(T^{\varepsilon} - T^0) - cT^{\varepsilon} \ln \frac{T^{\varepsilon}}{T^0}. \tag{9}$$

In Eq. (9), the displacement field is approximated by its first-order asymptotic expansion (6a), whereas the temperature field is taken as the zeroth-order approximation (6b), i.e.,

$$\begin{aligned} \boldsymbol{u}^{\varepsilon}(\boldsymbol{x}, \boldsymbol{y}) &= \boldsymbol{u}^{(0)}(\boldsymbol{x}) + \varepsilon\, \boldsymbol{u}^{(1)}(\boldsymbol{x}, \boldsymbol{y}) \\ &= \boldsymbol{u}^{(0)}(\boldsymbol{x}) + \varepsilon\, e_{xpq}(\boldsymbol{u}^{(0)})(\boldsymbol{x})\, \boldsymbol{N}^{pq}(\boldsymbol{y}) + \varepsilon\left(T^{(0)}(\boldsymbol{x}) - T^0\right) \boldsymbol{P}(\boldsymbol{y}), \end{aligned} \tag{10}$$

$$T^{\varepsilon}(\boldsymbol{x}, \boldsymbol{y}) = T^{(0)}(\boldsymbol{x}). \tag{11}$$

The corresponding strain tensor $e_{ij}(\boldsymbol{u}^{\varepsilon})$ decomposes into macroscopic and microscopic contributions:

$$\begin{aligned} e_{ij}(\boldsymbol{u}^{\varepsilon})(\boldsymbol{x}, \boldsymbol{y}) &= e_{xij}(\boldsymbol{u}^{(0)})(\boldsymbol{x}) + e_{yij}(\boldsymbol{u}^{(1)})(\boldsymbol{x}, \boldsymbol{y}) + \varepsilon\, e_{xij}(\boldsymbol{u}^{(1)})(\boldsymbol{x}, \boldsymbol{y}) \\ &= e_{xij}(\boldsymbol{u}^{(0)})(\boldsymbol{x}) + e_{xpq}(\boldsymbol{u}^{(0)})(\boldsymbol{x})\, e_{yij}(\boldsymbol{N}^{pq})(\boldsymbol{y}) \\ &\quad + \left(T^{(0)}(\boldsymbol{x}) - T^0\right) e_{yij}(\boldsymbol{P})(\boldsymbol{y}) \\ &\quad + \frac{\varepsilon}{2}\left[e_{xpqj}(\boldsymbol{u}^{(0)})(\boldsymbol{x})\, N_i^{pq}(\boldsymbol{y}) + e_{xpqi}(\boldsymbol{u}^{(0)})(\boldsymbol{x})\, N_j^{pq}(\boldsymbol{y})\right]. \end{aligned} \tag{12}$$

Here, the influence of macroscopic temperature gradients on microscopic strain tensor is neglected.

For convenience in the subsequent derivations, we introduce the following shorthand notation:

$$\mathcal{E}_{ij} := e_{xij}(\boldsymbol{u}^{(0)}), \quad \mathcal{E}_{ijk} := \frac{\partial \mathcal{E}_{ij}}{\partial x_k} = e_{xijk}(\boldsymbol{u}^{(0)}), \quad T := T^{(0)}, \quad \dot{\boldsymbol{u}} := \frac{\partial \boldsymbol{u}^{(0)}}{\partial t},$$
$$\dot{\mathcal{E}}_{ij} := \frac{\partial e_{xij}(\boldsymbol{u}^{(0)})}{\partial t},$$

which denote, respectively, the macroscopic strain, strain gradient, temperature, velocity, and strain rate. Furthermore, we use the notation

$$\langle f \rangle(\boldsymbol{x}) = \int_{Y^s} f(\boldsymbol{x}, \boldsymbol{y})\,\mathrm{d}\boldsymbol{y} \tag{13}$$

to represent the integration over the unit cell, of any function $f(\boldsymbol{x}, \boldsymbol{y})$ defined on the two-scale domain.

By substituting Eqs. (11) and (12) into Eq. (9), and subsequently performing the integration over the unit cell Y^s, we obtain the following volume-averaged free energy density

$$\begin{aligned}\langle w^{\mathrm{f}}_{\varepsilon}\rangle(\boldsymbol{x}) &= \int_{Y} w^{\mathrm{f}}_{\varepsilon}(\boldsymbol{x}, \boldsymbol{y})\,\mathrm{d}\boldsymbol{y} \\ &= \frac{1}{2}\hat{C}_{ijkl}\mathcal{E}_{ij}(\boldsymbol{x})\mathcal{E}_{kl}(\boldsymbol{x}) + \frac{\varepsilon^2}{2}\hat{D}_{ijklmn}\mathcal{E}_{ijk}(\boldsymbol{x})\mathcal{E}_{lmn}(\boldsymbol{x}) + \frac{1}{2}\hat{G}\left(T-T^0\right)^2(\boldsymbol{x}) \\ &\quad - \hat{\beta}_{ij}\mathcal{E}_{ij}(\boldsymbol{x})\left(T-T^0\right)(\boldsymbol{x}) + c\left(T-T^0\right)(\boldsymbol{x}) - cT(\boldsymbol{x})\ln\frac{T(\boldsymbol{x})}{T^0}.\end{aligned} \tag{14}$$

which constitutes a part of the macroscopic homogenized Helmholtz free energy density. In Eq. (14), the nonlocal terms $\frac{\varepsilon^2}{2}\hat{D}_{ijklmn}\mathcal{E}_{ijk}\mathcal{E}_{lmn} + \frac{1}{2}\hat{G}\left(T-T^0\right)^2$ represent additional stored energy associated with microscopic inhomogeneous deformation arising from microstructural heterogeneities. The effective (homogenized) coefficients are explicitly given by

$$\begin{cases} \hat{C}_{ijkl} = \displaystyle\int_{Y^s}\left[C_{ijkl} + C_{ijpq}\,e_{ypq}(\boldsymbol{N}^{kl})(\boldsymbol{y})\right]\mathrm{d}\boldsymbol{y}, & (15a) \\ \hat{D}_{ijklmn} = \displaystyle\int_{Y^s} C_{pkqn}\,N_p^{ij}(\boldsymbol{y})\,N_q^{lm}(\boldsymbol{y})\,\mathrm{d}\boldsymbol{y}, & (15b) \\ \hat{\beta}_{ij} = \displaystyle\int_{Y^s}\left[\beta\,\delta_{ij} + \beta\,e_{ykk}(\boldsymbol{N}^{ij})(\boldsymbol{y})\right]\mathrm{d}\boldsymbol{y}, & (15c) \\ \hat{G} = \displaystyle\int_{Y^s}\left[C_{ijkl}\,e_{yij}(\boldsymbol{P})(\boldsymbol{y})\,e_{ykl}(\boldsymbol{P})(\boldsymbol{y}) - 2\beta\,e_{ykk}(\boldsymbol{P})(\boldsymbol{y})\right]\mathrm{d}\boldsymbol{y}. & (15d) \end{cases}$$

The detailed derivation of Eq. (14) is provided in Appendix A.

In deriving the volume-averaged kinetic energy, thermal effects are neglected. Accordingly, the volume-averaged kinetic energy density can be expressed as (Rao et al., 2023):

$$\begin{aligned}\langle w^{\mathrm{k}}_{\varepsilon}\rangle(\boldsymbol{x}) &= \int_{Y} w^{\mathrm{k}}_{\varepsilon}(\boldsymbol{x}, \boldsymbol{y})\,\mathrm{d}\boldsymbol{y} \\ &= \tfrac{1}{2}\rho\,\dot{u}_i(\boldsymbol{x})\,\dot{u}_i(\boldsymbol{x}) + \tfrac{\varepsilon^2}{2}\,\hat{F}_{ijkl}\,\dot{\mathcal{E}}_{ij}(\boldsymbol{x})\,\dot{\mathcal{E}}_{kl}(\boldsymbol{x}),\end{aligned} \tag{16}$$

where the fourth-order micro-inertia tensor is defined as

$$\hat{F}_{ijkl} = \int_{Y^s} \rho\,N_p^{ij}(\boldsymbol{y})\,N_p^{kl}(\boldsymbol{y})\,\mathrm{d}\boldsymbol{y}. \tag{17}$$

In Eq. (16), the term $\frac{1}{2}\rho\,\dot{u}_i(\boldsymbol{x})\,\dot{u}_i(\boldsymbol{x})$ corresponds to the classical macroscopic kinetic energy density, while the additional contribution $\frac{\varepsilon^2}{2}\,\hat{F}_{ijkl}\,\dot{\mathcal{E}}_{ij}(\boldsymbol{x})\,\dot{\mathcal{E}}_{kl}(\boldsymbol{x})$ originates from the microscopic fluctuations of the displacement field. This higher-order term can be interpreted as a microstructural contribution to the macroscopic Helmholtz free energy storage of the medium.

The volume-averaged fracture energy density is obtained by averaging the total fracture energy over the reference cell Y_ε:

$$w^{\mathrm{frac}}(\boldsymbol{x}) = \frac{1}{|Y_\varepsilon|}\,G_c\,l(\boldsymbol{x}) = \frac{1}{\varepsilon}G_c\,d(\boldsymbol{x}), \tag{18}$$

where G_c denotes the critical fracture energy (or critical energy release rate). The homogenized fracture energy $w^{\mathrm{frac}}(\boldsymbol{x})$ is also part of the macroscopic Helmholtz free energy.

According to Eqs. (14), (16), and (18), the macroscopic homogenized Helmholtz free energy and kinetic energy are given by

$$\begin{cases} \begin{aligned} w^{\mathrm{f}} &= \langle w^{\mathrm{f}}_{\varepsilon}\rangle + w^{\mathrm{frac}} + \frac{\varepsilon^2}{2}\hat{F}_{ijkl}\dot{\mathcal{E}}_{ij}\dot{\mathcal{E}}_{kl} \\ &= \frac{1}{2}\hat{C}_{ijkl}\mathcal{E}_{ij}\mathcal{E}_{kl} + \frac{\varepsilon^2}{2}\hat{D}_{ijklmn}\mathcal{E}_{ijk}\mathcal{E}_{lmn} + \frac{1}{2}\hat{G}(T-T^0)^2 - \hat{\beta}_{ij}\mathcal{E}_{ij}(T-T^0) \\ &\quad + c(T-T^0) - cT\ln\frac{T}{T^0} + \frac{1}{\varepsilon}G_c d + \frac{\varepsilon^2}{2}\hat{F}_{ijkl}\dot{\mathcal{E}}_{ij}\dot{\mathcal{E}}_{kl}, \end{aligned} & (19a) \\ w^{\mathrm{k}} = \frac{1}{2}\rho\,\dot{u}_i\dot{u}_i. & (19b) \end{cases}$$

Remark 1. The nonlocal contributions associated with the effective tensors $\hat{\boldsymbol{D}}$, $\hat{\boldsymbol{G}}$, and $\hat{\boldsymbol{F}}$ characterize the influence of microscopic perturbations in the displacement, temperature, and velocity fields on the macroscopic energy storage. In contrast to many classical nonlocal damage models, where such higher-order terms are introduced phenomenologically at the macroscopic level, the present formulation derives them directly from the underlying microstructure through an asymptotic homogenization procedure.

Remark 2.

According to the definition of the homogenized energy functional, all homogenized coefficients depend on the normalized microcrack length d and its orientation θ (see Fig. 3). For a fixed crack length d, the angular dependence of the effective material parameters is fully determined by the orthogonal rotation tensor

$$\mathbf{R}(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

acting on the reference configuration $\theta = 0$. Specifically, one has

$$\begin{cases} \hat{C}_{ijkl}(d,\theta) = \hat{C}_{pqrs}(d,0)\,R_{ip}(\theta)R_{jq}(\theta)R_{kr}(\theta)R_{ls}(\theta), & (20a) \\ \hat{D}_{ijklmn}(d,\theta) = \hat{D}_{opqrst}(d,0)\,R_{io}(\theta)R_{jp}(\theta)R_{kq}(\theta)R_{lr}(\theta)R_{ms}(\theta)R_{nt}(\theta), & (20b) \\ \hat{F}_{ijkl}(d,\theta) = \hat{F}_{pqrs}(d,0)\,R_{ip}(\theta)R_{jq}(\theta)R_{kr}(\theta)R_{ls}(\theta), & (20c) \\ \hat{\beta}_{ij}(d,\theta) = \hat{\beta}_{pq}(d,0)\,R_{ip}(\theta)R_{jq}(\theta), & (20d) \\ \hat{G}(d,\theta) = \hat{G}(d,0), & (20e) \end{cases}$$

which maps the homogenized material parameters evaluated in the reference configuration ($\theta = 0$) to that corresponding to an arbitrarily oriented microcrack.

3. Macroscopic thermo–mechanical–damage framework

In this section, we formulate a coupled thermo–mechanical–damage model in accordance with the first and second laws of thermodynamics. The derivation is based on the macroscopic homogenized Helmholtz free energy density (19a) and kinetic energy density (19b).

3.1. Energy dissipation inequality

In this subsection, we establish the macroscopic energy dissipation inequality in accordance with the first and second laws of thermodynamics.

The first law of thermodynamics, incorporating strain-gradient effects, can be expressed in terms of the rate of the macroscopic internal energy density w^{in} as

$$\dot{w}^{\mathrm{in}} = \Sigma_{ij}\,\dot{\mathcal{E}}_{ij} + \Gamma_{ijk}\,\dot{\mathcal{E}}_{ijk} - \nabla\cdot\boldsymbol{Q}, \tag{21}$$

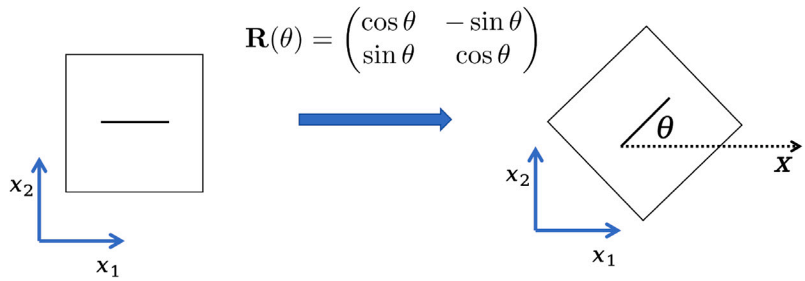


Fig. 3. Reference RVE ($\theta = 0$) and its rotated configuration obtained via $R(\theta)$.

where Σ_{ij} denotes the macroscopic Cauchy stress, Γ_{ijk} is the higher-order (double) stress conjugate to the strain gradient $\mathcal{E}_{ijk}$, and $\boldsymbol{Q}$ is the macroscopic heat flux.

The second law of thermodynamics, is expressed in the form of Clausius–Duhem inequality:

$$\dot{\eta} + \nabla \cdot \left(\frac{\boldsymbol{Q}}{T} \right) = \frac{1}{T} \left(T\dot{\eta} + \nabla \cdot \boldsymbol{Q} + \frac{\boldsymbol{Q} \cdot \nabla T}{T} \right) \geq 0, \tag{22}$$

where η is the macroscopic entropy density. The entropy density η and the internal energy density w^{in} are related to the macroscopic Helmholtz free energy density w^{f} through the thermodynamic relations

$$\begin{cases} \eta = \dfrac{\partial w^{\text{f}}}{\partial T}, & (23a) \\ w^{\text{in}} = w^{\text{f}} - T\eta & (23b) \\ \quad = \frac{1}{2}\hat{C}_{ijkl}\mathcal{E}_{ij}\mathcal{E}_{kl} + \frac{\varepsilon^2}{2}\hat{D}_{ijklmn}\mathcal{E}_{ijk}\mathcal{E}_{lmn} - \frac{1}{2}\hat{G}(T-T^0)(T+T^0) + \hat{\beta}_{ij}\mathcal{E}_{ij}T^0 & \\ \quad\quad + c\,(T-T^0) + \frac{1}{\varepsilon}G_c d + \frac{\varepsilon^2}{2}\hat{F}_{ijkl}\dot{\mathcal{E}}_{ij}\dot{\mathcal{E}}_{kl}. & \end{cases}$$

From these relations, one further obtains

$$\dot{w}^{\text{f}} = \dot{w}^{\text{in}} - T\dot{\eta} - \eta\dot{T}. \tag{24}$$

Combining the above expressions, the general form of the thermodynamic dissipation inequality is given by

$$\Sigma_{ij}\dot{\mathcal{E}}_{ij} + \Gamma_{ijk}\dot{\mathcal{E}}_{ijk} - \eta\dot{T} - \dot{w}^{\text{f}} - \frac{\boldsymbol{Q} \cdot \nabla T}{T} \geq 0. \tag{25}$$

By substituting the macroscopic Helmholtz free energy density (19a), the dissipation inequality can be expressed explicitly in terms of the macroscopic fields as

$$\begin{aligned} &\left[\Sigma_{ij} - \frac{\partial w^{\text{f}}}{\partial \mathcal{E}_{ij}} - \varepsilon^2 \hat{F}_{ijkl}\ddot{\mathcal{E}}_{kl} \right] \dot{\mathcal{E}}_{ij} \\ &+ \left[\Gamma_{ijk} - \frac{\partial w^{\text{f}}}{\partial \mathcal{E}_{ijk}} \right] \dot{\mathcal{E}}_{ijk} - \frac{\partial w^{\text{f}}}{\partial d}\dot{d} + \left[-\eta - \frac{\partial w^{\text{f}}}{\partial T} \right] \dot{T} + \frac{\boldsymbol{Q} \cdot \nabla T}{T} \geq 0. \end{aligned} \tag{26}$$

This inequality imposes a thermodynamic consistency constraint on the admissible constitutive relations, heat conduction and damage evolution.

3.2. Damage evolution law

To close the macroscopic problem, an evolution law for the damage variable, interpreted as the normalized micro-crack length $d(\boldsymbol{x}, t) \in [0, 1]$, is required. In the present work, we adopt a physically motivated modeling choice that links the damage rate to the homogenized energy release rate.

According to the dissipation inequality (26) and the irreversibility constraint of the damage process, the admissible evolution of d must satisfy

$$-\frac{\partial w^{\text{f}}}{\partial d}\dot{d} \geq 0, \qquad \dot{d} \geq 0. \tag{27}$$

which enforces the non-negativity of the damage driving force together with the monotonic increase of the damage variable.

Motivated by the classical fracture criteria of Griffith (1921) and Irwin (1958), crack propagation is assumed to be activated when the energy release rate exceeds the critical fracture energy G_c. For the homogenized setting, the energy release rate associated with the reference cell Y_ε around a macroscopic point $\boldsymbol{x}$ is given by

$$\begin{aligned} \mathcal{G} &= -\frac{\partial |Y_\varepsilon^s| (\langle w^{\text{f}} \rangle + \langle w^{\text{k}} \rangle)}{\partial l} = -\varepsilon^2 \frac{\partial (\langle w^{\text{f}} \rangle + \langle w^{\text{k}} \rangle)}{\partial l} = -\varepsilon \frac{\partial (\langle w^{\text{f}} \rangle + \langle w^{\text{k}} \rangle)}{\partial d} \\ &= -\frac{\varepsilon}{2}\frac{\partial \hat{C}_{ijkl}}{\partial d}\mathcal{E}_{ij}\mathcal{E}_{kl} - \frac{\varepsilon^3}{2}\frac{\partial \hat{D}_{ijklmn}}{\partial d}\mathcal{E}_{ijk}\mathcal{E}_{lmn} + \varepsilon\frac{\partial \hat{\beta}_{ij}}{\partial d}\mathcal{E}_{ij}(T-T^0) \\ &\quad - \frac{\varepsilon}{2}\frac{\partial \hat{G}}{\partial d}(T-T^0)^2 - \frac{\varepsilon^3}{2}\frac{\partial \hat{F}_{ijkl}}{\partial d}\dot{\mathcal{E}}_{ij}\dot{\mathcal{E}}_{kl}. \end{aligned} \tag{28}$$

When $\mathcal{G} > G_c$, the microcrack becomes unstable and propagates. Building upon the dynamic fracture framework of Freund (1990), the corresponding damage evolution law is expressed as

$$\dot{d} = \frac{1}{\varepsilon}\dot{l} = \begin{cases} \dfrac{1}{\varepsilon}C_R\left(1 - \dfrac{G_c}{\mathcal{G}}\right), & \text{if } \mathcal{G} > G_c, \\ 0, & \text{if } \mathcal{G} \leq G_c, \end{cases} \tag{29}$$

where C_R denotes the Rayleigh wave speed, introduced here as a characteristic upper-bound velocity scale for crack advance.

By combining Eqs. (28) and (29), it can be seen that microcrack propagation is jointly driven by the macroscopic strain, strain gradient, strain rate, and temperature fields. According to our previous work (Rao et al., 2023), an increase in the macroscopic strain promotes crack growth, whereas higher strain gradients and strain rates tend to suppress it. This behavior reflects the inherent resistance of the microstructure to highly localized deformation and rapid temporal variations. The gradient and rate terms thus provide a stabilizing effect by elevating the energetic barrier to fracture. In the present work, we further incorporate the influence of temperature on microcrack evolution through the homogenized thermal modulus $\hat{\beta}_{ij}$ and the nonlocal parameter $\hat{G}$, which will be analyzed in detail in the following sections.

It remains to verify the thermodynamic consistency of (29). For $\mathcal{G} \leq G_c$, one has $\dot{d} = 0$, and condition (27) is trivially satisfied. For $\mathcal{G} > G_c$, the free energy derivative with respect to d is

$$\frac{\partial w^{\text{f}}}{\partial d} = \frac{1}{\varepsilon}\left(-\mathcal{G} + G_c\right), \tag{30}$$

such that

$$-\frac{\partial w^{\text{f}}}{\partial d}\dot{d} = \frac{C_R}{\varepsilon^2}\frac{(\mathcal{G} - G_c)^2}{\mathcal{G}} \geq 0. \tag{31}$$

Therefore, the proposed damage evolution law not only captures the dependence of micro-crack growth on macroscopic strain, strain gradients, loading rate effects, and thermal fields, but also ensures full consistency with the thermodynamic energy dissipation inequality.

Remark 3. Within this two-scale framework, microcracks are not explicitly tracked. Instead, their collective effects are homogenized through the representative unit cell with periodic boundary conditions, yielding a diffuse damage field: a macroscopically continuous yet spatially localized field that effectively captures the averaged behavior of microcrack ensembles. This approach allows the macroscopic model to account for microscale thermomechanical effects, including stress concentration at microcrack tips, displacement discontinuities along crack surfaces, and thermal dissipation during crack propagation. The diffuse nature of the damage field also provides a natural regularization that prevents pathological localization and ensures numerical stability.

3.3. Thermo–mechanical coupled equations

Starting from the energy dissipation inequality, the thermodynamically consistent constitutive relations can be derived:

$$\begin{cases} \Sigma_{ij} = \dfrac{\partial w^{\mathrm{f}}}{\partial \mathcal{E}_{ij}} + \varepsilon^2 \hat{F}_{ijkl}\ddot{\mathcal{E}}_{kl} = \hat{C}_{ijkl}\mathcal{E}_{kl} - \hat{\beta}_{ij}(T-T^0) + \varepsilon^2 \hat{F}_{ijkl}\ddot{\mathcal{E}}_{kl}, & \text{(32a)} \\ \Gamma_{ijk} = \dfrac{\partial w^{\mathrm{f}}}{\partial \mathcal{E}_{ijk}} = \varepsilon^2 \hat{D}_{ijklmn}\mathcal{E}_{lmn}. & \text{(32b)} \end{cases}$$

Here, Σ_{ij} and Γ_{ijk} denote the homogenized stress and higher-order stress associated with the strain gradient, respectively.

The balance of linear momentum at the macroscopic scale, which accounts for the strain gradient effect, takes the form (Lam et al., 2003):

$$\rho\,\ddot{u}_i - \frac{\partial \Sigma_{ij}}{\partial x_j} + \frac{\partial^2 \Gamma_{ijk}}{\partial x_j \partial x_k} = f_i. \tag{33}$$

For the variation of internal energy in the material, two equivalent expressions can be obtained. On the one hand, according to the first law of thermodynamics (Eq. (21)) together with the constitutive relations (Eq. (32)), we have

$$\begin{aligned} \dot{w}^{\mathrm{in}} &= \Sigma_{ij}\dot{\mathcal{E}}_{ij} + \Gamma_{ijk}\dot{\mathcal{E}}_{ijk} - \nabla\cdot\boldsymbol{Q} + r \\ &= \hat{C}_{ijkl}\mathcal{E}_{kl}\dot{\mathcal{E}}_{ij} - \hat{\beta}_{ij}(T-T^0)\dot{\mathcal{E}}_{ij} + \varepsilon^2 \hat{F}_{ijkl}\ddot{\mathcal{E}}_{kl}\dot{\mathcal{E}}_{ij} + \varepsilon^2 \hat{D}_{ijklmn}\mathcal{E}_{lmn}\dot{\mathcal{E}}_{ijk} - \nabla\cdot\boldsymbol{Q} + r. \end{aligned} \tag{34}$$

On the other hand, from the macroscopic expression of the internal energy (Eq. (23b)), it follows that

$$\begin{aligned} \dot{w}^{\mathrm{in}} &= \hat{C}_{ijkl}\mathcal{E}_{kl}\dot{\mathcal{E}}_{ij} + \varepsilon^2 \hat{D}_{ijklmn}\mathcal{E}_{lmn}\dot{\mathcal{E}}_{ijk} - \hat{G}T\dot{T} + \hat{\beta}_{ij}\dot{\mathcal{E}}_{ij}T^0 + c\dot{T} \\ &\quad + \varepsilon^2 \hat{F}_{ijkl}\ddot{\mathcal{E}}_{kl}\dot{\mathcal{E}}_{ij} + \frac{\partial w^{\mathrm{in}}}{\partial d}\dot{d}. \end{aligned} \tag{35}$$

Therefore, since the right-hand sides of Eqs. (34) and (35) are identical, eliminating the common terms yields the macroscopic heat conduction equation:

$$c\dot{T} - \hat{G}T\dot{T} + \hat{\beta}_{ij}\dot{\mathcal{E}}_{ij}T = -\nabla\cdot\boldsymbol{Q} + r - \frac{\partial w^{\mathrm{in}}}{\partial d}\dot{d}. \tag{36}$$

The additional source term $-\frac{\partial w^{\mathrm{in}}}{\partial d}\dot{d}$ accounts for the residual energy release associated with crack growth, i.e., the difference between the internal energy reduction due to damage and the fracture energy expended in creating new crack surfaces. The damage rate $\dot{d}$ controls the amount of energy converted to heat.

To close the model, Fourier's law is homogenized at the macroscale. At the microscale, the local heat flux within the periodic cell Y is given by

$$q_i(\boldsymbol{x},\boldsymbol{y}) = -\kappa\left(\frac{\partial T^{(0)}}{\partial x_i} + \frac{\partial M_j(\boldsymbol{y})}{\partial y_i}\frac{\partial T^{(0)}}{\partial x_j}\right), \tag{37}$$

and averaging over Y yields the effective flux

$$Q_i(\boldsymbol{x}) = \int_Y q_i(\boldsymbol{x},\boldsymbol{y})\,\mathrm{d}\boldsymbol{y} = -\hat{\kappa}_{ij}\frac{\partial T}{\partial x_j}, \tag{38}$$

where the homogenized conductivity tensor reads

$$\hat{\kappa}_{ij} = \int_Y \kappa\left(\delta_{ij} + \frac{\partial M_j}{\partial y_i}\right)\mathrm{d}\boldsymbol{y}. \tag{39}$$

The homogenized thermal conductivity tensor $\hat{\kappa}_{ij}$ inherits its dependence on the normalized microcrack length d and crack orientation θ through the first-order cell function M_j, thereby reflecting the anisotropic degradation of heat-transport capacity induced by microcrack evolution. For a fixed crack length d, the angular dependence also follows the standard tensorial rotation rule,

$$\hat{\kappa}_{ij}(d,\theta) = \hat{\kappa}_{pq}(d,0)\,R_{ip}(\theta)\,R_{jq}(\theta). \tag{40}$$

Finally, substituting $\boldsymbol{Q}$ into the macroscopic energy balance leads to the thermo–damage conduction equation:

$$c\dot{T} - \hat{G}T\dot{T} + \hat{\beta}_{ij}\dot{\mathcal{E}}_{ij}T = \frac{\partial}{\partial x_i}\left(\hat{\kappa}_{ij}\frac{\partial T}{\partial x_j}\right) + r - \frac{\partial w^{\mathrm{in}}}{\partial d}\dot{d}. \tag{41}$$

As demonstrated in Appendix B, the homogenized thermal conductivity tensor $\hat{\kappa}_{ij}$ is positive semi-definite. This property ensures that the homogenized Fourier's law automatically satisfies the constraint imposed by the dissipation inequality Eq. (26), namely

$$-\frac{\boldsymbol{Q}\cdot\nabla T}{T} = \frac{1}{T}\left(\hat{\kappa}_{ij}\frac{\partial T}{\partial x_i}\frac{\partial T}{\partial x_j}\right) \geq 0. \tag{42}$$

3.4. Summary

In summary, the coupled thermo–mechanical–damage model incorporating the strain-gradient effect can be expressed as

$$\begin{cases} \rho\,\ddot{u}_i - \dfrac{\partial}{\partial x_j}\left(\hat{C}_{ijkl}\,\mathcal{E}_{kl} - \hat{\beta}_{ij}(T-T^0) + \varepsilon^2 \hat{F}_{ijkl}\,\ddot{\mathcal{E}}_{kl}\right) + \varepsilon^2\dfrac{\partial^2}{\partial x_j \partial x_k}\left(\hat{D}_{ijklmn}\,\mathcal{E}_{lmn}\right) = f_i, & \text{(43a)} \\ c\,\dot{T} - \hat{G}\,T\dot{T} + \hat{\beta}_{ij}\,\dot{\mathcal{E}}_{ij}\,T = \dfrac{\partial}{\partial x_i}\left(\hat{\kappa}_{ij}\dfrac{\partial T}{\partial x_j}\right) + r - \dfrac{\partial w^{\mathrm{in}}}{\partial d}\,\dot{d}, & \text{(43b)} \\ \dot{d} = \begin{cases} \dfrac{1}{\varepsilon}\,C_R\left(1 - \dfrac{G_c}{\mathcal{G}}\right), & \text{if } \mathcal{G} > G_c, \\ 0, & \text{if } \mathcal{G} \leq G_c. \end{cases} & \text{(43c)} \end{cases}$$

Moreover, Eqs. (32), (42), (23a), and (31) collectively verify that the proposed model satisfies the thermodynamic dissipation inequality, Eq. (26), thereby ensuring its thermodynamic consistency.

Remark 4 (*Multiscale Structure of the Proposed Model*)**.** The present thermo–mechanical–damage model (43) is based on an energetic equivalence established after truncating the asymptotic expansions of the displacement and temperature fields with respect to the small parameter ε. The truncated fields are substituted into the microscopic free energy density, which is then averaged over the fast variable $\boldsymbol{y}$ to obtain the macroscopic free energy density. A schematic illustration of the coupled macro–micro framework is presented in Fig. 4.

An alternative macro–micro strategy consists in first substituting the full asymptotic expansions of the microscopic fields into the microscopic free energy density, followed by integration over the RVE. This procedure yields an asymptotic expansion of the homogenized free energy density in powers of ε. In this case, additional $\mathcal{O}(\varepsilon^2)$ contributions arise from the second-order displacement expansion and the corresponding second-order cell functions.

Previous studies in the context of purely dynamic multiscale formulations have demonstrated that the present strategy, based on a first-order truncation, provides an effective compromise between predictive accuracy and computational efficiency, while avoiding the considerable algebraic and numerical complexity associated with second-order cell functions (Li et al., 2025b).

4. Numerical simulation

This section presents numerical simulations carried out to validate the proposed thermomechanical fracture model. First, the relationship between the homogenized material parameters and the normalized crack length d is examined. Next, the local macroscopic response is

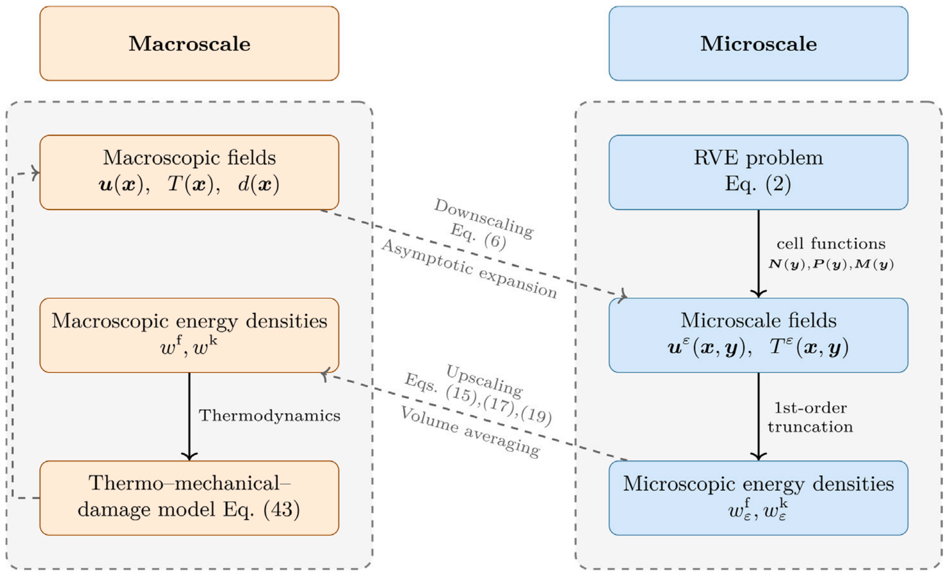


Fig. 4. Schematic of the coupled macro–micro framework, illustrating the downscaling reconstruction of microscopic fields from macroscopic variables via cell functions and the homogenization-based upscaling of the effective energy density.

Table 1
Material parameters of PMMA.

E (GPa)	ν	ρ (kg/m^3)	c (J/(m^3 K))	α (1/K)	κ (W/(m K))	C_R (m/s)	G_c (J/m^2)
5.5	0.36	1190	1190 · 1470	10^{-4}	0.2	1243	400

analyzed, with particular emphasis on the manifestation of non-local effects and thermal dissipation during a uniaxial tensile test. Finally, a tensile simulation of a V-notch plate and a dynamic impact test on a plate specimen are conducted, the latter being compared with experimentally measured temperature to further assess the model's applicability and accuracy.

The material employed in the simulations is polymethyl methacrylate (PMMA), and the corresponding material parameters, obtained from experimental data, are summarized in Table 1. These parameters, taken from Bjerke and Lambros (2003) and Dascalu and Gbetchi (2019), serve as the foundation for the numerical analyses presented herein.

4.1. Homogenized material parameters

In this section, we investigate the homogenized parameters of the coupled thermo–mechanical–damage model, which are defined as integrals involving the first-order cell functions $\boldsymbol{N}^{ij}$, $\boldsymbol{P}$, and $\boldsymbol{M}$, and thus depend nonlinearly on the normalized microcrack length d and orientation θ.

To evaluate these coefficients, we first fix the microcrack orientation at $\theta = 0$ and compute the numerical solutions of the cell functions by solving the governing PDEs, Eqs. (8a)–(8c), using the finite element method for a discrete set of d values, $d = \{0, 0.025, 0.05, \ldots, 0.99, 0.995, 0.99999\}$. For each d, the homogenized material parameters are directly obtained from their definitions, Eqs. (15), (39). Continuous representations of the coefficients are then constructed via cubic spline interpolation to ensure smooth and continuous curves. Finally, the coefficients for arbitrary d and θ are obtained using the rotation formula, Eq. (20). Some representative numerical results are presented in Fig. 5. Further analyses of the cell function $\boldsymbol{N}$ and the homogenized coefficients $\hat{C}_{ijkl}$, $\hat{D}_{ijklmn}$, and $\hat{F}_{ijkl}$ for $\theta = 0$ are discussed in Rao et al. (2022, 2023).

Fig. 5(a) shows the numerical solutions of the cell functions $\boldsymbol{P}$ and $\boldsymbol{M}$ for $\theta = 0$ and $d = 0.5$. It is observed that $\boldsymbol{P}$ and $\boldsymbol{M}$ exhibit distinct symmetry and antisymmetry characteristics along directions parallel and perpendicular to the crack, consistent with the theoretical predictions given in Eqs. (A.1) and (A.2). These symmetry properties are critical, as they directly influence the anisotropic behavior of the homogenized coefficients.

Fig. 5(b) illustrate the effects of microcracks on homogenized thermal properties and nonlocal parameters with $\theta = 0$. The thermal moduli $\hat{\beta}_{11}$ and $\hat{\beta}_{22}$ decrease with increasing microcrack length, with $\hat{\beta}_{22}$ (normal to the crack) declining more sharply, reflecting anisotropic weakening of thermal expansion, while $\hat{\beta}_{11}$ (tangential) decreases more slowly. Similarly, the thermal conductivity $\hat{\kappa}_{22}$ is significantly reduced perpendicular to the crack, whereas $\hat{\kappa}_{11}$ remains nearly unchanged, highlighting strongly directional thermal transport.

The absolute value of the nonlocal parameter $\hat{G}$ increases monotonically with increasing damage, indicating progressively enhanced nonlocal effects. From Eqs. (28) and (29), the temperature field influences microcrack growth via $\partial\hat{\beta}_{ij}/\partial d < 0$ (stabilizing) and $\partial\hat{G}/\partial d < 0$ (destabilizing). Since the destabilizing effect of $\hat{G}$ is quadratic in ΔT, while $\hat{\beta}_{ij}$ is linear, elevated temperatures promote softening and accelerate microcrack propagation, consistent with experimental observations (Feng et al., 2019). Material-specific variations and strong coupling with the strain field, however, can lead to highly nonlinear thermo–mechanical responses.

We further examine the influence of the normalized microcrack length d and orientation θ on the macroscopic homogenized material coefficients. Fig. 5(c) present the resulting elastic stiffness and effective thermal conductivity. In particular, we focus on the material coefficients corresponding to a fully penetrated microcrack, i.e., $d = 1$. Both tensors exhibit pronounced anisotropy induced by the microcrack orientation. For elasticity, C_{1111} reaches its maximum when the crack aligns with x_1 ($\theta = 0°$) and its minimum at $\theta = 90°$, whereas C_{2222} exhibits the opposite trend. Shear components C_{1112} and C_{1212} attain their peaks near $\theta = 45°$ and $135°$ due to mixed-mode (opening–sliding) deformation. Thermal conductivity shows a similar pattern: κ_{11} is largest along the crack and smallest perpendicular to it, while κ_{22} exhibits the reverse behavior. Off-diagonal terms κ_{12} vanish at

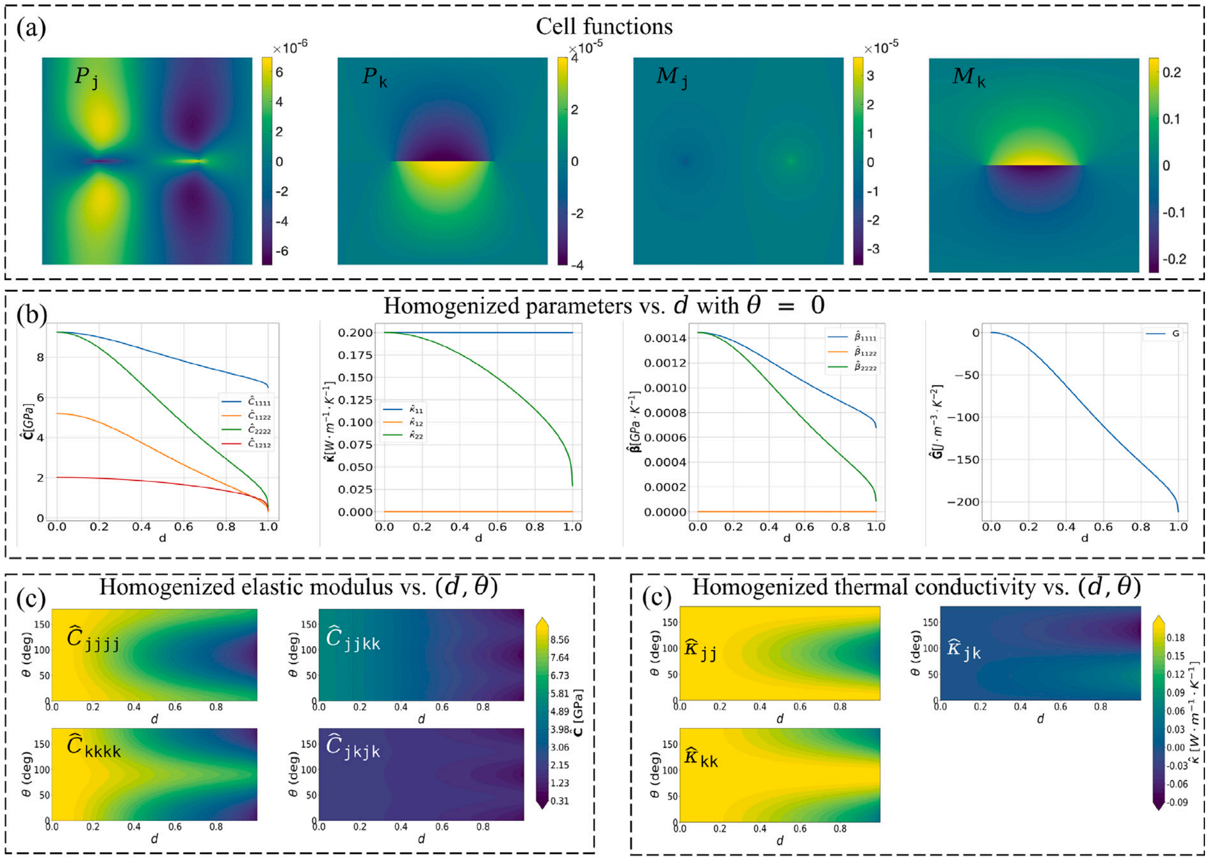


Fig. 5. Computation of homogenized coefficients: (a) Cell functions from Eqs.,(8a), (8b), and (8c); (b) Homogenized coefficients at $\theta = 0$ from Eqs., (15), (39); (c) Homogenized coefficients for arbitrary θ and d from Eq.,(20).

$\theta = 0°$ and 90°, reaching extrema at oblique angles, consistent with the tensor rotation law, Eq. (40). Overall, microcracks induce a coherent anisotropy: stiffness and thermal transport along the normal direction are most significantly degraded, whereas properties along the crack-parallel direction remain comparatively robust. Shear and off-diagonal components attain their maxima at oblique orientations due to enhanced mixed-mode effects. These trends corroborate both the physical fidelity and the internal consistency of the multiscale homogenization framework.

Based on the above observations, and by combining the expression of the energy release rate (Eq. (28)), it follows that for Mode-I dominated fracture, microcracks whose planes are normal to the principal tensile stress are most readily activated and driven to extend. Although microcracks in actual heterogeneous materials exhibit statistically random orientations, the energetically preferred propagation direction remains aligned with the opening-mode driving force. Therefore, in the subsequent analysis we focus on the representative configuration in which the principal tensile stress acts in the vertical direction, while the microcrack is oriented horizontally.

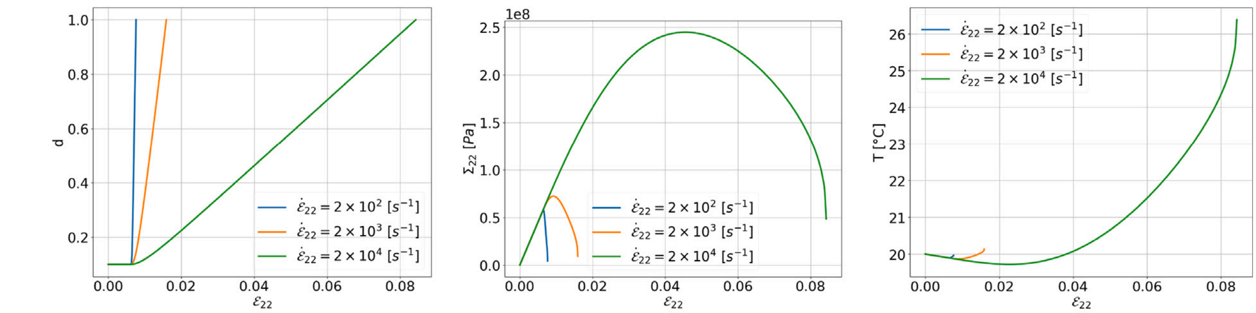
4.2. Local analysis

In this subsection, we analyze the local mechanical responses of a macroscopic material point, specifically at a Gauss point in finite element simulations. We examine the evolution of crack length, temperature, and the constitutive response under various loading conditions. For this analysis, we set $\dot{\mathcal{E}}_{11} = \dot{\mathcal{E}}_{12} = 0$, corresponding to a uniaxial tensile test with the strain gradient tensor $\mathcal{E}_{ijk} = 0$, as detailed in Rao et al. (2023), and assume adiabatic conditions at high strain rates. Accordingly, the temperature $\mathcal{T}$ is computed using Eq. (43b), while the damage variable is determined using Eq. (43c), with the initial temperature and damage set to $T^0 = 20\,°$Celsius and $d_0 = 0.1$, respectively.

First, we fix the characteristic microstructural length $\varepsilon = 5 \times 10^{-5}$ m and vary the applied strain rate. Fig. 6 shows the influence of strain rate on the evolution of the physical fields during loading. As shown in Fig. 6(a), at low strain rates, micro-crack propagation exhibits a brittle response, i.e., once instability initiates, the crack rapidly spans the entire RVE. As the strain rate increases, crack propagation becomes more ductile, and the time interval from instability onset to complete penetration increases. Fig. 6(b) indicates that the constitutive response similarly transitions from brittle to ductile with increasing strain rate. At low strain rates, the stress increases linearly with loading and drops sharply upon reaching the peak stress. In contrast, at higher strain rates, stress growth is initially linear but gradually slows down before reaching the maximum, followed by a gradual decrease.

Fig. 6(c) shows that, under loading, the temperature initially decreases slightly. When crack propagation begins, the release of internal energy due to crack growth dominates over the energy absorbed in creating new surfaces and other dissipative mechanisms, leading to a rapid temperature rise. Both the rate and magnitude of the temperature increase become larger as the strain rate increases.

Finally, Fig. 7 summarizes the peak stress (i.e., fracture strength) and peak temperature after complete crack penetration as a function of strain rates. The results show that the tensile strength increases with strain rate, exhibiting an abrupt rise following a plateau regime. This suggests that the material can sustain a higher level of damage prior to catastrophic failure at elevated strain rates, in agreement with experimental observations (Lambert and Ross, 2000; Zinszner et al., 2015). Concurrently, the associated temperature rise becomes more pronounced, also consistent with reported measurements (Bjerke and Lambros, 2002). Previous studies (Wang and Ramesh, 2004; Zhou et al.,



(a) Damage d **(b)** Stress Σ_{22} **(c)** Temperature T

Fig. 6. Strain-rate-dependent evolution of local physical quantities.

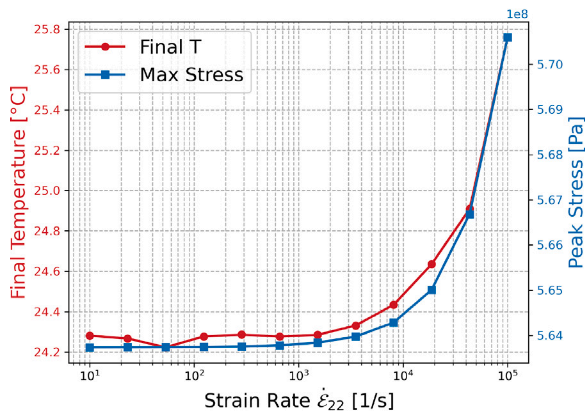


Fig. 7. Effect of strain rate on final temperature and peak stress.

2006; Li et al., 2018) have shown that dynamic loading promotes extensive microcracking, producing finer fragments than those observed at lower strain rates. Such mechanisms not only enhance the tensile strength but also contribute to increased thermal fracture dissipation.

Secondly, the strain rate is fixed at $3\times10^3\ \mathrm{m}^{-1}$, while the microstructural length-scale parameter ε is varied. Fig. 8 illustrates the influence of ε on the evolution of the physical fields during loading. As shown in Figs. 8(a) and 8(b), decreasing ε delays the onset of damage and enhances the maximum stress that the material can sustain; a consistent trend is also observed in Fig. 9.

Furthermore, Figs. 8(c) and 9 indicate that the temperature rise at complete failure is more pronounced for smaller values of ε.

This behavior can be rationalized as follows. A smaller ε corresponds to a higher density of smaller microcracks. In this case, the fracture process involves the progressive activation and coalescence of numerous microcracks, rather than the rapid propagation of fewer large cracks. Consequently, the energy dissipation is distributed over a larger number of microcracking events, which delays macroscopic damage initiation and increases the load-bearing capacity. At the same time, the cumulative energy release during the dense microcrack interactions is more effectively converted into heat, leading to a larger temperature rise at failure.

4.3. Macro-structure analysis

Based on the dependence of the homogenized material parameters on the damage field d established in Section 4.1, we further analyze brittle fracture in macro-structures together with the associated thermal effects of crack propagation. To this end, the coupled thermo–mechanical–damage system (43) is solved by employing the staggered algorithm described in Algorithm 1. In this procedure, the macroscopic momentum balance (43a) and the heat conduction Eq. (43b) are discretized by the finite element method. Triangular meshes are adopted, with P_2 Lagrangian elements used for displacement and temperature interpolation to properly capture strain gradient effects, while piecewise constant P_0 elements are employed for the damage field.

Algorithm 1: Staggered algorithm at time interval $[t_n, t_{n+1}]$

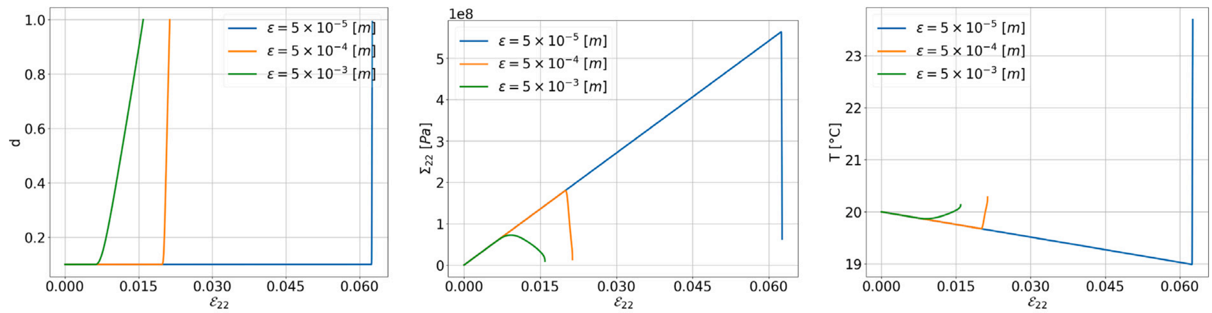
Input: Solutions of the temperature field T_n, displacement field $\boldsymbol{u}_n$, and damage field d_n at time t_n
Output: Updated fields T_{n+1}, $\boldsymbol{u}_{n+1}$, and d_{n+1} at time t_{n+1}

Set iteration counter $i = 0$, tolerance $\varepsilon = 10^{-8}$;
Initialize $T^{(0)}_{n+1} \leftarrow T_n$, $\boldsymbol{u}^{(0)}_{n+1} \leftarrow \boldsymbol{u}_n$, $d^{(0)}_{n+1} \leftarrow d_n$;
repeat
  Update homogenized material coefficients and their damage derivatives based on $d^{(i)}_{n+1}$;
  Compute $T^{(i+1)}_{n+1}, \boldsymbol{u}^{(i+1)}_{n+1}$ while fixing $d^{(i)}_{n+1}$;
  if $d^{(i+1)}_{n+1} < 1$ **then**
    Compute energy release rate $G^{(i+1)}$;
    Update $d^{(i+1)}_{n+1}$ while fixing $T^{(i+1)}_{n+1}, \boldsymbol{u}^{(i+1)}_{n+1}$;
  else
    Set $d^{(i+1)}_{n+1} \leftarrow 1$;
  end
  $i \leftarrow i + 1$;
until $|T^{(i+1)}_{n+1} - T^{(i)}_{n+1}| + \|\boldsymbol{u}^{(i+1)}_{n+1} - \boldsymbol{u}^{(i)}_{n+1}\| + |d^{(i+1)}_{n+1} - d^{(i)}_{n+1}| \le \varepsilon$;
Update solutions: $T_{n+1} \leftarrow T^{(i)}_{n+1}$, $\boldsymbol{u}_{n+1} \leftarrow \boldsymbol{u}^{(i)}_{n+1}$, $d_{n+1} \leftarrow d^{(i)}_{n+1}$;

4.3.1. Tensile fracture of a V-notched plate

In this subsection, we present the simulation of Mode-I fracture for a rectangular PMMA plate with dimensions $0.1\ \mathrm{m} \times 0.05\ \mathrm{m}$, containing a V-notch of depth $b = 0.01\ \mathrm{m}$ and opening angle $\beta = \arctan(1/5)$, as illustrated in Fig. 10(a). Owing to geometric symmetry about the central axis, only half of the domain is modeled using a symmetric finite-element mesh, with local refinement introduced in the region expected for crack propagation (Fig. 10(b)).

In the first group of simulations, the normalized microcrack length (initial damage) is set to $d_0 = 0.1$, and the mean microcrack spacing is $\varepsilon = 1 \times 10^{-5}\ \mathrm{m}$. The time-integration step is $\Delta t = 1 \times 10^{-7}\ \mathrm{s}$. The loading velocity is varied to investigate the rate-dependent fracture behavior. Fig. 11 shows the evolution of the crack and temperature fields under a tensile velocity of $40\ \mathrm{m/s}$. As the loading proceeds, the main crack initiates from the V-notch tip and propagates toward the right boundary. Distinct crack branching is observed, accompanied by the formation of multiple secondary cracks near the main crack path. A pronounced temperature rise develops at the tips of both the main and secondary cracks. This heating effect remains highly localized, with negligible temperature changes outside the crack region. Such thermal localization is consistent with experimental observations reported by Fuller et al. (1975), Weichert and Schönert (1978).



(a) Damage d **(b)** Stress Σ_{22} **(c)** Temperature T

Fig. 8. ε-dependent evolution of local physical quantities.

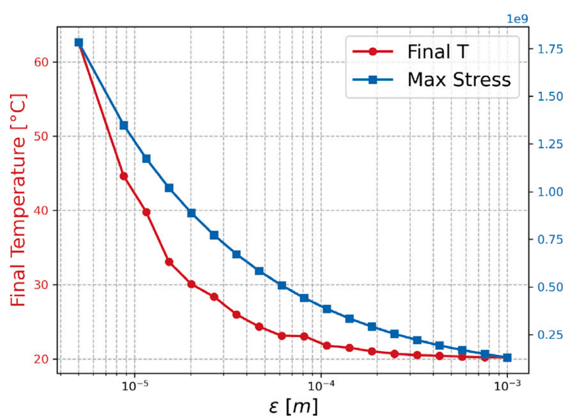


Fig. 9. Effect of ε on final temperature and peak stress.

Figs. 12(a) and 12(b) respectively show the temperature evolution at a point located 10 mm ahead of the crack tip and the spatially averaged temperature over the specimen. Both curves confirm that the temperature rise is strongly localized: in the present simulations, the local temperature increase exceeds 8 K, whereas the average temperature rise remains around 0.5 K. As shown in Fig. 12(a), the magnitude of the local temperature rise increases with the loading rate, in agreement with the local thermomechanical analysis in Section 4.2. Moreover, Fig. 12(b) reveals an initial cooling stage before crack initiation, which also aligns with experimental measurements (Fuller et al., 1975; Bougaut and Rittel, 2001).

As further evidenced in Fig. 12, increasing the loading rate leads to larger displacements loading when the crack tip reaches a given position, implying greater deformation and thus a higher resistance to crack opening. This trend is also reflected in Fig. 12(c), which presents the spatially average stress–displacement curves for three different loading velocities. The results show that both the peak (critical) stress and the overall ductility increase with the loading rate, consistent with the rate-dependent responses discussed in Section 4.2. This behavior can be attributed to the kinetics of energy barriers and the inertia effects in the vicinity of the advancing crack front (Reinhardt and Weerheijm, 1991).

We next examine the influence of the microstructural length scale ε with $v = 10\,\mathrm{m/s}$ and an initial damage variable $d_0 = 0.1$. The time-integration step is $\Delta t = 1\times10^{-7}$ s.

Fig. 13 displays the fracture zone and the distributions of temperature fields at complete fracture for different ε values. As ε decreases, a pronounced intensification of the local temperature rise is observed. This behavior can be attributed to the increase in microcrack density at smaller ε, which promotes more frequent microcrack interactions and consequently enhances energy dissipation within the fracture zone.

Figs. 14(a) and 14(b) further show the temporal evolution of the local temperature at a point located 10 mm ahead of the crack tip and the specimen-averaged temperature, respectively. The results consistently demonstrate that a smaller ε leads to a higher temperature rise at complete failure. Together with Fig. 13, these findings confirm that the heat generation induced by crack propagation remains highly localized, confined to the vicinity of the crack paths.

Moreover, Figs. 14(a,b) reveal that decreasing ε enhances the material's resistance to deformation. This tendency is also reflected in Fig. 14(c), which presents the average stress–displacement curves for various ε values. The results indicate that the critical (peak) stress increases as ε decreases, implying that smaller microstructural length scales yield a higher macroscopic load-carrying capacity at failure. In contrast to the loading-rate-dependent case, however, the ductility exhibits only a weak dependence on ε. These observations are consistent with the local thermomechanical analyses discussed in Section 4.2.

4.3.2. Impact fracture of a rectangular notched plate

In this subsection, numerical simulations of Mode-I dynamic fracture in PMMA are carried out and compared with the experimental data reported in Bjerke and Lambros (2002, 2003). The experimental configuration consists of single-edge notched specimens with dimensions 292 mm in height, 127 mm in length, and 8.5 mm in thickness. A starter notch of 25 mm is introduced at one edge to initiate fracture. Dynamic Mode-I fracture is triggered by impacting the opposite edge with a cylindrical steel projectile of length 75 mm and diameter 38 mm, as illustrated in Fig. 15(a).

The infrared detector array employed to detect local temperature changes consists of 16 square HgCdTe photovoltaic elements, each with a side length of 80 μm, arranged linearly with a spacing of 20 μm. Detector No. 8 is aligned with the notch tip at the onset of fracture. The numerical simulations primarily focus on the temperature evolution at the center of detector No. 8, which is then compared against the experimentally measured temperature histories at detectors No. 8, 9, and 10.

In the finite element simulations, a symmetric mesh is employed with local refinement in the anticipated crack propagation zone, as depicted in Fig. 15(b). Since the lateral dimensions of the specimen are much larger than its thickness, the deformation is reasonably approximated by a plane strain state (Xiang et al., 2021). In the experiments, traction-free boundaries were approximated by supporting the specimen on cantilevered glass slides (Bjerke and Lambros, 2002).

In the first test, the projectile impacts the specimen at a velocity of 25 m/s. The initial damage field is set to $d = 0.1$, and the average microcrack spacing is $\varepsilon = 1\times10^{-5}$ mm. As shown in Fig. 16, three dominant cracks initiate from both the impact surface and the notch tip, and subsequently propagate with time. The crack emanating from the notch tip extends horizontally across the specimen and intersects the detector array, with multiple secondary branches developing along the path.

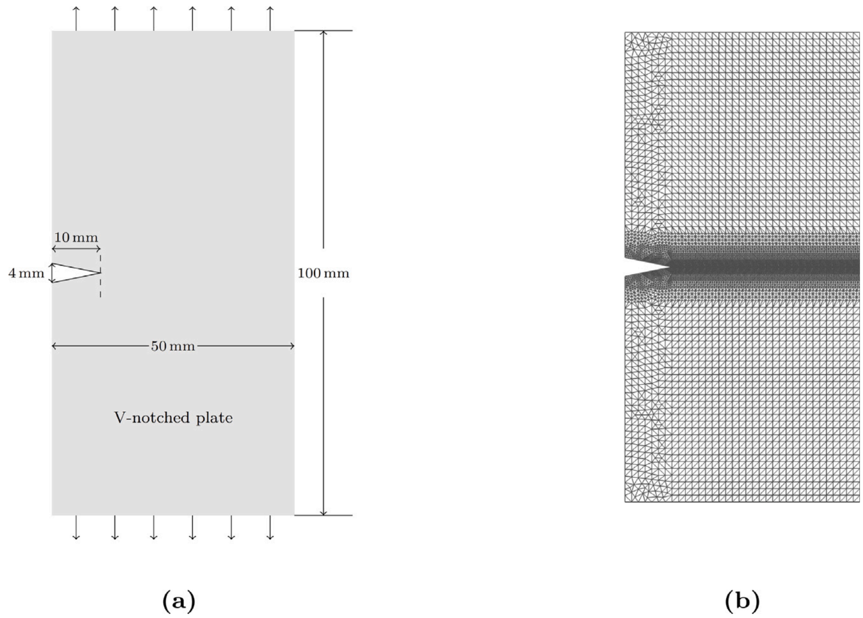


Fig. 10. Tensile fracture of a V-notch plate:(a) geometry and boundary conditions; (b) finite element mesh.

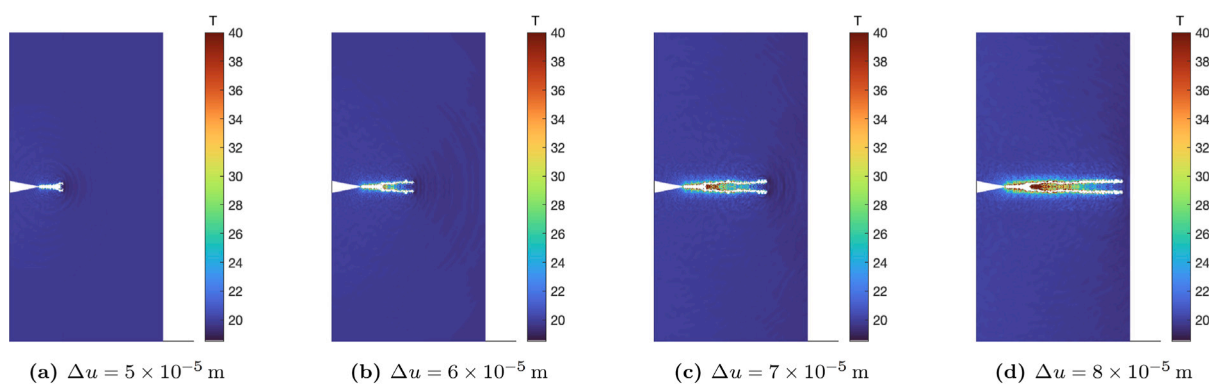


Fig. 11. Evolution of fracture zones and temperature fields at different loading stages.

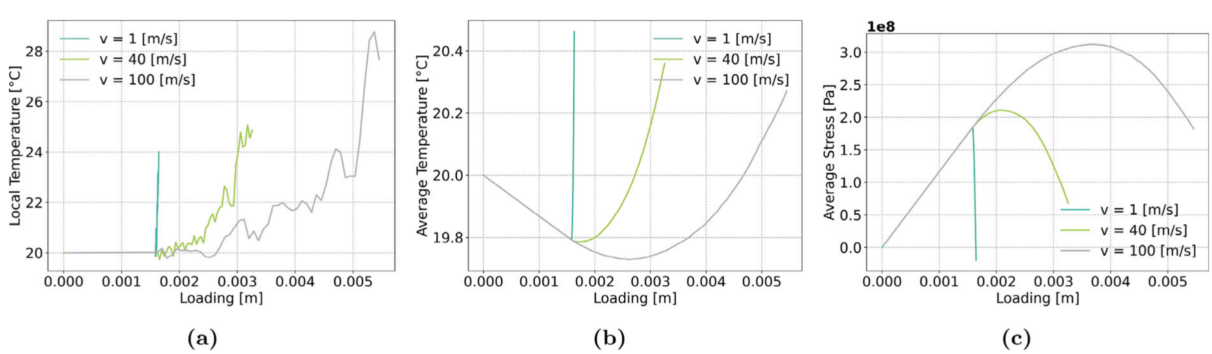
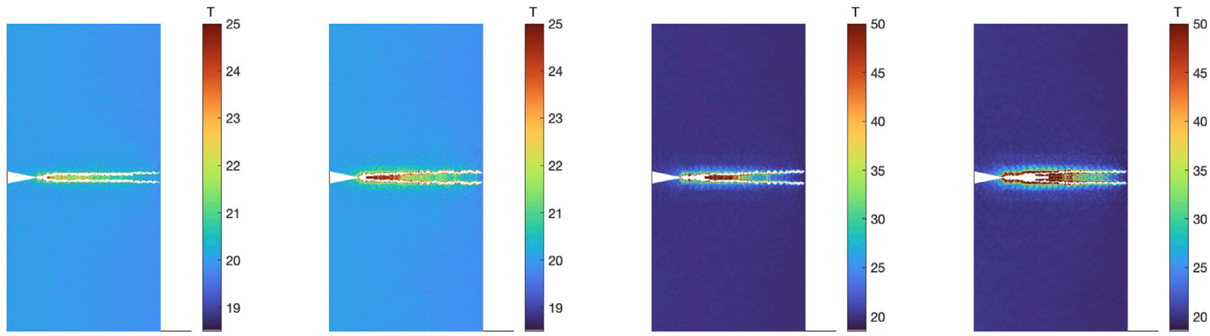


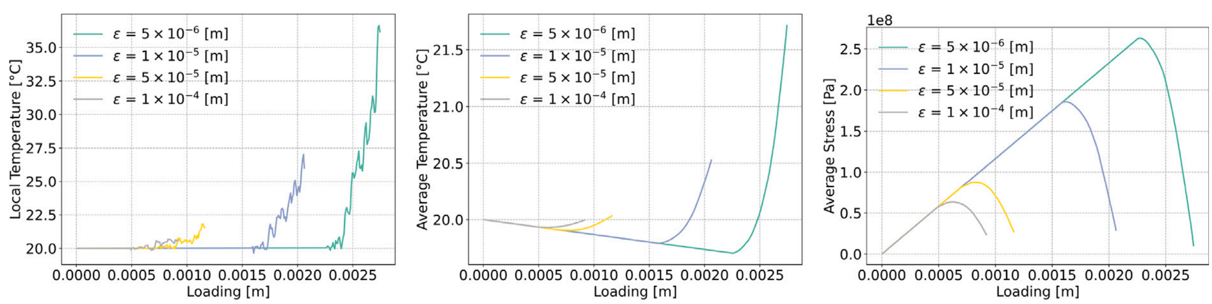
Fig. 12. Evolution of temperature and stress fields under different loading velocities: (a) local temperature; (b) spatially averaged temperature; (c) spatially averaged stress.

A pronounced temperature rise is observed at the tips of the main and secondary cracks during propagation. Fig. 17 further shows the temperature changes ΔT at the center of detector No. 8. Both experimental and numerical results indicate that the temperature rise becomes evident at approximately 270 μs to 280 μs, with an average increase of about 10 °C. The numerical predictions capture both the magnitude and temporal trend of the experimental data with good accuracy. The remaining discrepancy may be attributed to frictional effects within the material and heat transfer to the surroundings, which are not accounted for in the model. In addition, the actual microcracks in the material exhibit more complex sizes and orientations than the simplified assumption in this work. Moreover, in the experiments, once the crack surfaces separate, the detector increasingly observes the air gap between the crack faces rather than the hot crack surfaces themselves, resulting in the post-peak decline in temperature observed experimentally (Bjerke and Lambros, 2002, 2003).



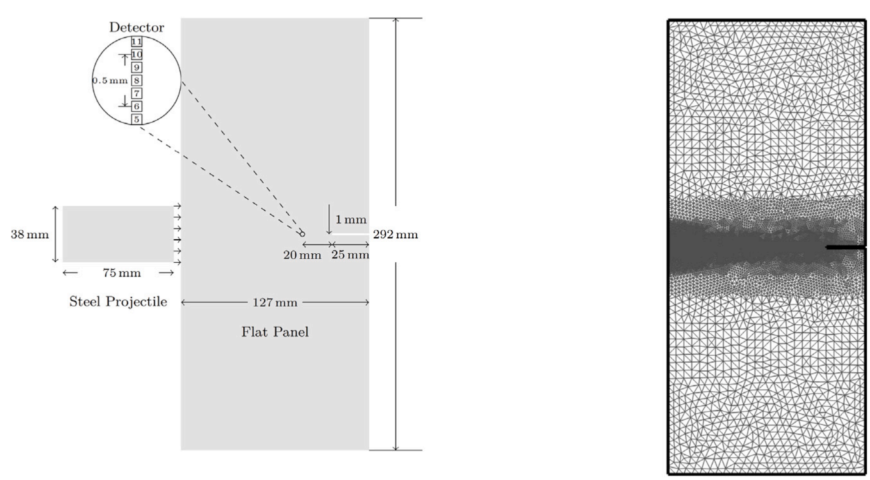
(a) $\varepsilon = 1 \times 10^{-4}$ m (b) $\varepsilon = 5 \times 10^{-5}$ m (c) $\varepsilon = 1 \times 10^{-5}$ m (d) $\varepsilon = 5 \times 10^{-6}$ m

Fig. 13. The final fracture zones and temperature fields for different ε.



(a) (b) (c)

Fig. 14. Evolution of temperature and stress fields for different values of the length-scale parameter ε: (a) local temperature; (b) spatially averaged temperature; (c) spatially averaged stress.



(a) (b)

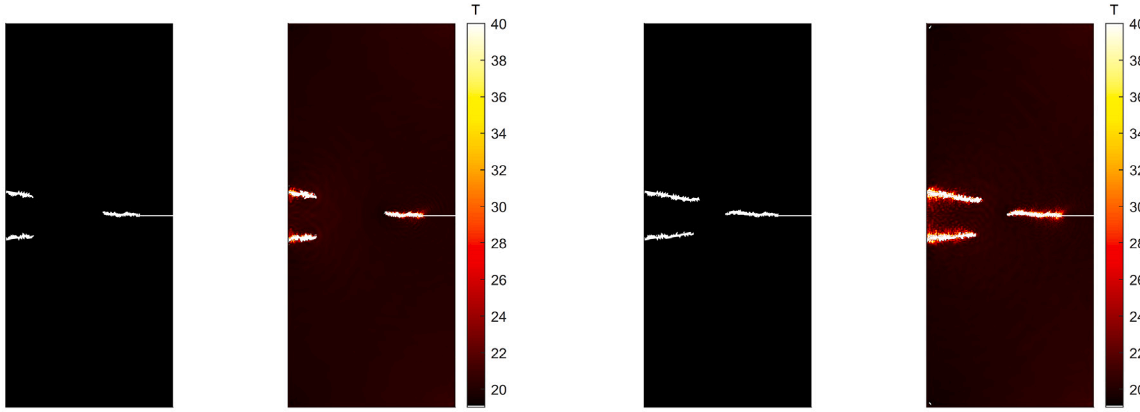
Fig. 15. Impact fracture of a rectangular notched plate:(a) geometry and boundary conditions; (b) finite element mesh.

In the second test, the projectile velocity is increased to 40 m/s, with the same initial damage field (d = 0.1) and micro-crack spacing ($\varepsilon = 1 \times 10^{-5}$ mm). As shown in Fig. 18, the overall crack pattern remains similar to that in the first test, consisting of three dominant cracks and numerous secondary branches. However, due to the higher impact velocity, the cracks propagate significantly faster: comparable crack lengths are reached within a much shorter time. This effect is also reflected in the temperature response. As shown in Fig. 19, a distinct temperature rise near the detector array occurs as early as 110 μs, in contrast to the delayed rise at approximately 270 μs in Fig. 17.

Moreover, the magnitude of the local temperature rise increases with impact velocity. In the second test, the average temperature rise reaches about 30 °C (Fig. 19), compared with only 10 °C in the first test (Fig. 17). Again, the comparison between numerical and experimental results demonstrates good agreement, both in the onset time of the temperature rise and in its magnitude.

4.3.3. Thermo–mechanical fracture of a rectangular notched plate

In this section, the fracture response of a rectangular notched plate subjected to coupled thermo–mechanical loading is systematically analyzed. The geometric configuration and associated boundary conditions are illustrated in Fig. 20(a), whereas the corresponding finite element discretization is shown in Fig. 20(b). A displacement-controlled tensile load is applied on the top and bottom boundaries in incremental steps



(a) Fracture zone at 300 µs **(b)** Temperature at 300 µs **(c)** Fracture zone at 350 µs **(d)** Temperature at 350 µs

Fig. 16. Fracture evolution and associated temperature fields in the panel under 25 m/s impact (Experiment 1).

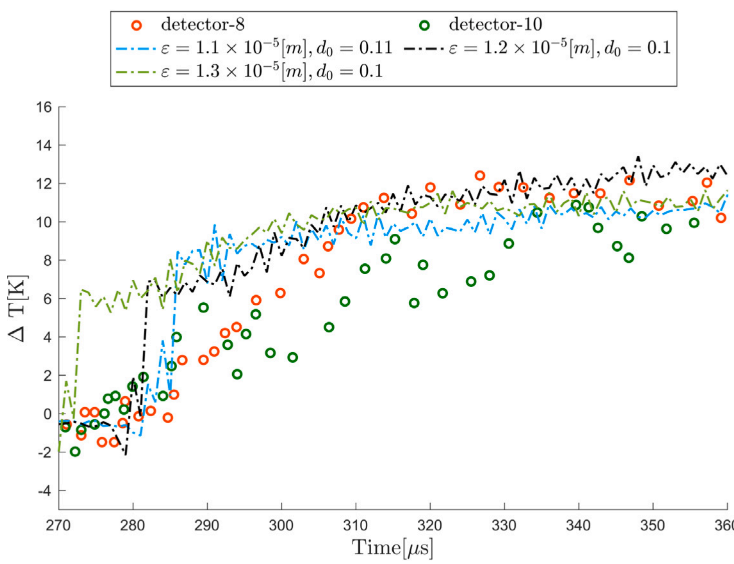
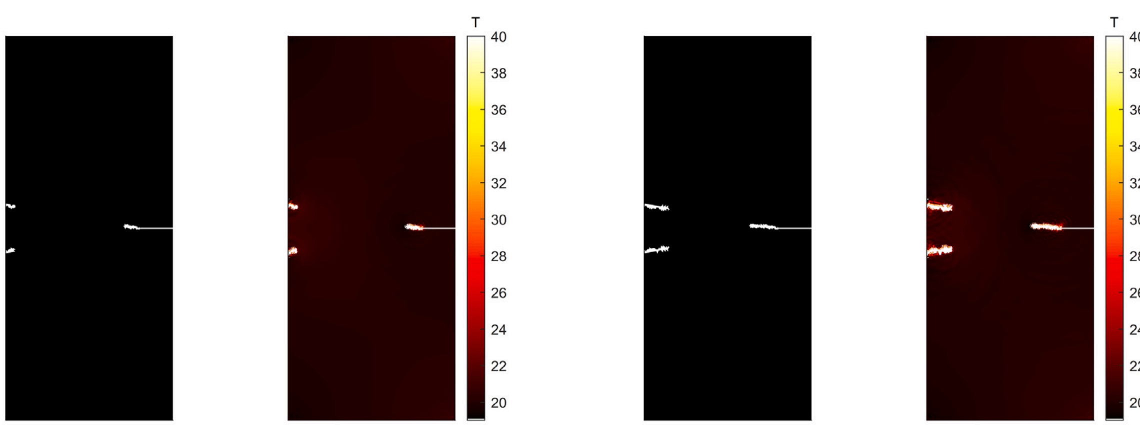


Fig. 17. Comparison of simulated and experimental temperature evolution (Bjerke and Lambros, 2003) under 25 m/s impact.



(a) Fracture zone at 130 µs **(b)** Temperature at 130 µs **(c)** Fracture zone at 150 µs **(d)** Temperature at 150 µs

Fig. 18. Fracture evolution and associated temperature fields in the panel under 40 m/s impact (Experiment 2).

of $\Delta u = 1{\times}10^{-3}$ mm. Simultaneously, the bottom boundary is maintained at a constant reference temperature $T_0 = 293$ K, while a prescribed thermal loading rate $\dot{T}$ is imposed on the top boundary to induce thermally driven expansion.

A representative crack evolution under coupled thermo–mechanical loading is depicted in Fig. 20(c), corresponding to quasi-static thermal loading with incremental temperature steps of $\Delta T = 0.1$ K, eventually

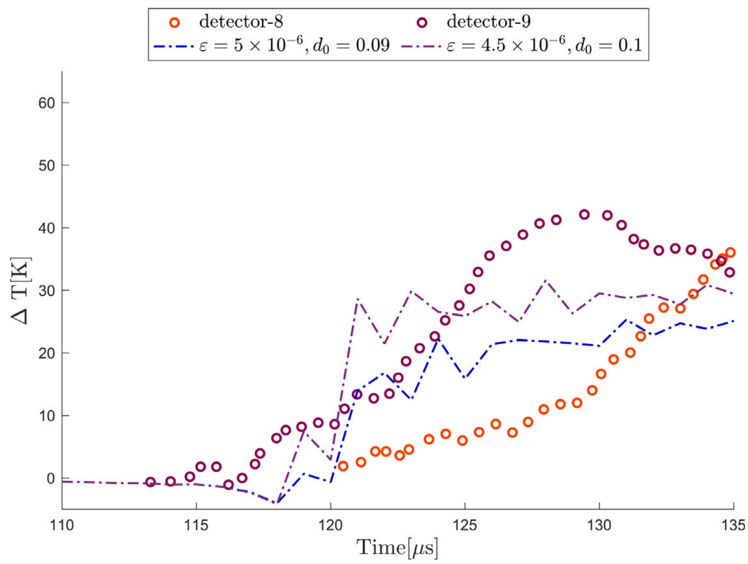


Fig. 19. Comparison of simulated and experimental temperature evolution (Bjerke and Lambros, 2003) under 40 m/s impact.

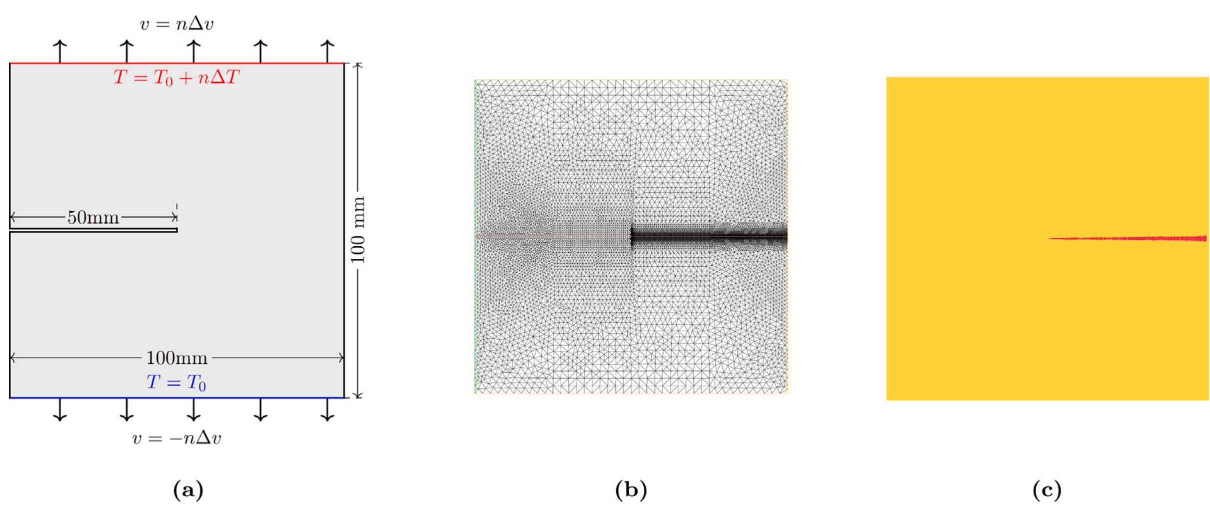


Fig. 20. Thermo–mechanical fracture of a rectangular notched plate: (a) geometry and boundary conditions; (b) finite element mesh; and (c) crack propagation under $\Delta T = 0.1$ K.

reaching a total temperature increase of approximately 50 K, with a thermal expansion coefficient of $\alpha = 1 \times 10^{-3}\ \mathrm{K}^{-1}$.

The present investigation emphasizes the role of key thermal parameters, including the thermal loading rate, the coefficient of thermal expansion, and the thermal diffusivity, in governing the fracture response. Here, the thermal diffusivity, defined as $D_T = \kappa/(\rho c_p)$, quantifies the rate at which heat propagates within the material. Particular attention is paid to the quantitative evaluation of crack initiation time and effective fracture strength under various thermo–mechanical loading conditions.

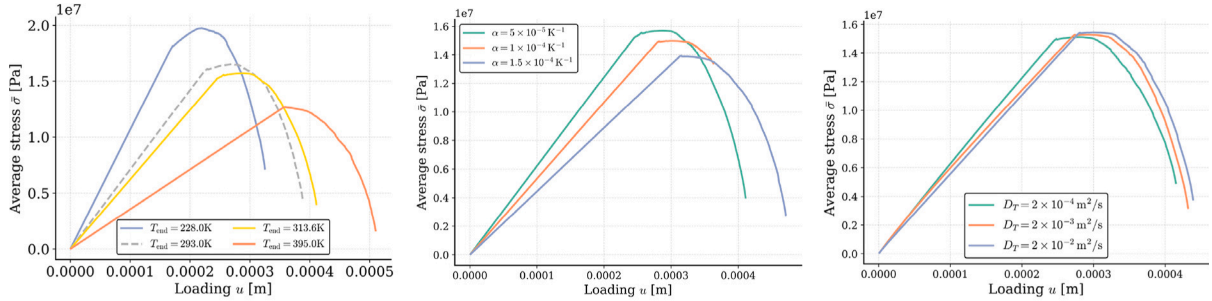
We first consider a quasi-static thermal loading regime, where the imposed thermal rate is negligible compared with the intrinsic thermal diffusivity D_T of the material, allowing the neglect of heat diffusion effects. Two parametric studies are conducted: (i) variation of the thermal loading step ΔT at a fixed thermal expansion coefficient $\alpha = 1{\times}10^{-4}\ \mathrm{K}^{-1}$, and (ii) variation of α for a fixed thermal loading increment $\Delta T = 0.1$ K. The resulting averaged stress–displacement curves are presented in Figs. 21(a) and 21(b), respectively, where T_{end} denotes the total temperature increase, given by the product of ΔT and the number of thermal loading steps.

As shown in Fig. 21(a), increasing the thermal loading amplitude ΔT systematically delays crack initiation and reduces the peak stress. Similarly, Fig. 21(b) demonstrates that increasing α produces a comparable effect, lowering the peak stress and postponing crack onset.

These trends can be interpreted within a unified thermo–mechanical framework, which may be understood from two complementary perspectives. First, under displacement-controlled loading, the total strain is prescribed. An increase in thermal loading (either ΔT or α) enhances the thermal strain contribution, thereby reducing the mechanical strain component. As a result, the local stress concentration at the notch tip is alleviated. From an energetic standpoint, the reduction in effective elastic strain leads to a decrease in the energy release rate, thereby weakening the fracture driving force and delaying crack initiation.

On the other hand, to attain the same critical energy release rate G_c, an increase in temperature implies a greater contribution from thermal expansion to the total energy. Consequently, a lower level of mechanical stress is required to satisfy the fracture criterion. This manifests macroscopically as a reduction in the apparent material strength under elevated thermal loading.

Subsequently, the effect of thermal diffusion is examined. In contrast to the quasi-static regime, here D_T is sufficiently low such that



(a) (b) (c)

Fig. 21. Averaged stress as a function of displacement under varying (a) temperature increment ΔT, (b) thermal expansion coefficient α, and (c) thermal diffusivity $D_T = \kappa/(\rho c_p)$.

non-uniform temperature fields may develop during loading. In the simulations, the thermal loading rate and the coefficient of thermal expansion are fixed at $\dot{T} = 1\times10^2$ K s^{-1} and $\alpha = 1\times10^{-4}$ K^{-1}, respectively, while the displacement loading corresponds to $\dot{v} = 1$ mm s^{-1}. Thermal diffusivity D_T is varied indirectly through adjustments in the thermal conductivity κ.

As illustrated in Fig. 21(c), a reduction in D_T intensifies thermal localization, accelerating both crack initiation and propagation, while only marginally reducing the peak stress. This behavior arises from the interplay between heat diffusion and localized deformation: lower D_T limits thermal redistribution, generating localized temperature gradients that produce heterogeneous thermal strains. These localized strains amplify incompatibilities and promote damage concentration, thereby accelerating fracture. Nevertheless, under predominantly displacement-controlled loading, the overall stress level is only slightly affected by variations in D_T, leaving the macroscopic peak stress largely unchanged.

5. Conclusion

This study develops a thermodynamically consistent two-scale nonlocal thermo–elastic–damage model for brittle materials containing microcracks. Starting from a representative volume element with an embedded microcrack, the homogenized Helmholtz free and kinetic energy densities are derived through a two-scale asymptotic expansion of the displacement and temperature fields, thereby incorporating the effects of microstructural length, strain gradient, strain rate, and temperature.

By applying the first and second laws of thermodynamics to the homogenized energy densities, an energy dissipation inequality is obtained, which guides the formulation of a higher-order constitutive model. The ensuing momentum and energy balance equations yield a set of coupled thermo–mechanical governing equations that account for strain gradient effects. In the heat conduction equation, the irreversible portion of internal energy dissipation exceeding the fracture energy for new surface formation is converted into heat. Combining these energy densities with Griffith's criterion leads to a microcrack evolution law governed by the macroscopic thermomechanical state variables (strain, strain gradient, strain rate, and temperature). The framework captures the conversion of dissipated energy at advancing crack tips into heat, while the nonlinear dependence of effective material parameters on microcrack length and orientation represents the anisotropic degradation of stiffness and conductivity. The model is shown to strictly satisfy the thermodynamic dissipation inequality, ensuring energetic consistency.

Local analyses reveal that the degradation of effective material properties is anisotropic, being most severe in the direction normal to the crack plane. The fracture behavior exhibits pronounced strain-rate dependence: as the applied strain rate increases, both the fracture strength and the local temperature rise increase, with stronger sensitivity observed at higher rates. Conversely, increasing the microstructural length scale ε reduces the microcrack density, thereby lowering both fracture strength and localized temperature rise.

Numerical simulations of tensile-induced fracture on V-notched plates further confirm these strain-rate and length-scale sensitivities, and demonstrate that the proposed model can predict crack propagation patterns, associated temperature fields, and their rate-dependent characteristics. In addition, simulations of impact-induced fracture show that the predicted temperature evolution is in excellent agreement with the experimental measurements reported in Bjerke and Lambros (2002), thereby validating the model's predictive capability for thermomechanically coupled brittle fracture. Furthermore, the thermo–mechanical fracture of rectangular notched plates is systematically analyzed to quantify the influence of loading rate and the sensitivity to key thermal parameters, including the thermal expansion coefficient and thermal diffusivity, in comparison with purely strain- and rate-dependent cases.

While the diffuse damage representation effectively captures the effects of individual microcracks, such as stiffness degradation and thermal dissipation associated with crack propagation, the current framework does not explicitly resolve discrete microcrack interactions, including branching and the coalescence of microcracks into macroscopic crack paths. These aspects will be addressed in future work.

CRediT authorship contribution statement

Quanzhang Li: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis. **Yipeng Rao:** Writing – review & editing, Validation, Software, Formal analysis. **Junzhi Cui:** Writing – review & editing, Validation, Supervision, Conceptualization. **Meizhen Xiang:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by National Natural Science Foundation of China (No. 12272061), National Natural Science Foundation of China (No. U2530210) and National Key Research and Development Program of China (No. 2024YFA1012101).

Appendix A. Homogenization of the RVE free energy

In this section, we derive the homogenized expression of the RVE free energy. Before proceeding with the analysis, we briefly recall the symmetry and antisymmetry properties of the cell functions. The symmetry of the displacement-type cell functions $\boldsymbol{N}^{pq}$ has been established in Rao et al. (2022) and will not be repeated here. For the first-order temperature-related cell functions, the following relations hold:

$$\begin{cases} P_1(y_1,y_2) = -P_1(1-y_1,y_2) = P_1(y_1,1-y_2), \\ P_2(y_1,y_2) = P_2(1-y_1,y_2) = -P_2(y_1,1-y_2), \end{cases} \tag{A.1}$$

$$\begin{cases} M_1(y_1,y_2) = M_1(y_1,1-y_2) = -M_1(1-y_1,y_2), \\ M_2(y_1,y_2) = -M_2(y_1,1-y_2) = M_2(1-y_1,y_2). \end{cases} \tag{A.2}$$

Substituting the asymptotic expansions of the strain field (12) and the temperature field (11) into the local free energy density (9), we obtain

$$\begin{aligned} w^f_\varepsilon(\boldsymbol{x},\boldsymbol{y}) =& \frac{1}{2}C_{ijkl}e_{ij}(\boldsymbol{u}^\varepsilon)e_{kl}(\boldsymbol{u}^\varepsilon) - \beta\delta_{ij}e_{ij}(\boldsymbol{u}^\varepsilon)(T^\varepsilon - T^0) + c(T^\varepsilon - T^0) - cT^\varepsilon \ln\frac{T^\varepsilon}{T^0} \\ =& \frac{1}{2}C_{ijkl}\left[\mathcal{E}_{ij}(\boldsymbol{x}) + \mathcal{E}_{pq}(\boldsymbol{x})e_{yij}(N^{pq})(\boldsymbol{y}) + \varepsilon\mathcal{E}_{pqj}(\boldsymbol{x})N^{pq}_i(\boldsymbol{y})\right] \\ &\cdot\left[\mathcal{E}_{kl}(\boldsymbol{x}) + \mathcal{E}_{pq}(\boldsymbol{x})e_{ykl}(N^{pq})(\boldsymbol{y}) + \varepsilon\mathcal{E}_{pql}(\boldsymbol{x})N^{pq}_k(\boldsymbol{y})\right] \\ &+ C_{ijkl}\left[\mathcal{E}_{ij}(\boldsymbol{x}) + \mathcal{E}_{pq}(\boldsymbol{x})e_{yij}(N^{pq})(\boldsymbol{y})\right]\left[(T-T^0)(\boldsymbol{x})e_{ykl}(\boldsymbol{P})(\boldsymbol{y})\right] \\ &+ C_{ijkl}\left[\varepsilon\mathcal{E}_{pqj}(\boldsymbol{x})N^{pq}_i(\boldsymbol{y})\right]\left[(T-T^0)(\boldsymbol{x})e_{ykl}(\boldsymbol{P})(\boldsymbol{y})\right] \\ &+ \frac{1}{2}C_{ijkl}\left[(T-T^0)(\boldsymbol{x})e_{yij}(\boldsymbol{P})(\boldsymbol{y})\right]\left[(T-T^0)(\boldsymbol{x})e_{ykl}(\boldsymbol{P})(\boldsymbol{y})\right] \\ &- \beta\delta_{ij}\left[\mathcal{E}_{ij}(\boldsymbol{x}) + \mathcal{E}_{pq}(\boldsymbol{x})e_{yij}(N^{pq})(\boldsymbol{y})\right](T-T^0)(\boldsymbol{x}) \\ &- \beta\delta_{ij}\left[(T-T^0)(\boldsymbol{x})e_{yij}(\boldsymbol{P})(\boldsymbol{y})\right](T-T^0)(\boldsymbol{x}) \\ &- \beta\delta_{ij}\left[\varepsilon\mathcal{E}_{pqj}(\boldsymbol{x})N^{pq}_i(\boldsymbol{y})\right](T-T^0)(\boldsymbol{x}) \\ &+ c(T-T^0)(\boldsymbol{x}) - cT^{(0)}\ln\frac{T^{(0)}}{T^0}(\boldsymbol{x}). \end{aligned} \tag{A.3}$$

Following our previous work (Rao et al., 2022), the first term, upon integration over Y_s with respect to the microscopic variable y, is given by

$$\frac{1}{2}\hat{C}_{ijkl}\mathcal{E}_{ij}(\boldsymbol{x})\mathcal{E}_{kl}(\boldsymbol{x}) + \frac{\varepsilon^2}{2}\hat{D}_{ijklmn}\mathcal{E}_{ijk}(\boldsymbol{x})\mathcal{E}_{lmn}(\boldsymbol{x}). \tag{A.4}$$

The second term vanishes upon integration, owing to the fact that $\boldsymbol{P} \in \mathcal{D}(Y^s)$ and by invoking the weak form of the cell problem associated with $\boldsymbol{N}^{pq}$.

By virtue of the symmetry and antisymmetry properties of the cell functions, we obtain

$$\int_{Y^s} C_{ijkl}\left[\varepsilon\mathcal{E}_{pqj}(\boldsymbol{x})N^{pq}_i(\boldsymbol{y})\right]\left[(T-T^0)(\boldsymbol{x})e_{ykl}(\boldsymbol{P})(\boldsymbol{y})\right]\mathrm{d}\boldsymbol{y} = 0, \tag{A.5}$$

and similarly,

$$\int_{Y^s} -\beta\delta_{ij}\left[\varepsilon\mathcal{E}_{pqj}(\boldsymbol{x})N^{pq}_i(\boldsymbol{y})\right](T-T^0)(\boldsymbol{x})\mathrm{d}\boldsymbol{y} = 0. \tag{A.6}$$

For the remaining contributions, one also has

$$\begin{aligned} &\int_{Y^s} \frac{1}{2}C_{ijkl}\left[(T-T^0)(\boldsymbol{x})e_{yij}(\boldsymbol{P})(\boldsymbol{y})\right]\left[(T-T^0)(\boldsymbol{x})e_{ykl}(\boldsymbol{P})(\boldsymbol{y})\right] \\ &\quad - \beta\delta_{ij}\left[(T-T^0)(\boldsymbol{x})e_{yij}(\boldsymbol{P})(\boldsymbol{y})\right](T-T^0)(\boldsymbol{x})\mathrm{d}\boldsymbol{y} \\ =& \frac{1}{2}\hat{G}\left(T-T^0\right)^2(\boldsymbol{x}). \end{aligned} \tag{A.7}$$

Furthermore,

$$\begin{aligned} &\int_{Y^s} -\beta\delta_{ij}\left[\mathcal{E}_{ij}(\boldsymbol{x}) + \mathcal{E}_{pq}(\boldsymbol{x})e_{yij}(N^{pq})(\boldsymbol{y})\right](T-T^0)(\boldsymbol{x})\mathrm{d}\boldsymbol{y} \\ =& -\int_{Y^s}\left[\beta\delta_{ij} + \beta e_{ykk}(N^{ij})(\boldsymbol{y})\right]\mathrm{d}\boldsymbol{y}\ \mathcal{E}_{ij}(\boldsymbol{x})(T-T^0)(\boldsymbol{x}) \\ =& -\hat{\beta}_{ij}\mathcal{E}_{ij}(\boldsymbol{x})(T-T^0)(\boldsymbol{x}). \end{aligned} \tag{A.8}$$

Finally,

$$\int_{Y^s} c(T-T^0)(\boldsymbol{x}) - cT^{(0)}\ln\frac{T^{(0)}}{T^0}(\boldsymbol{x})\mathrm{d}\boldsymbol{y} = c(T-T^0)(\boldsymbol{x}) - cT^{(0)}\ln\frac{T^{(0)}}{T^0}(\boldsymbol{x}). \tag{A.9}$$

Collecting all the above results, the averaged free energy is given by

$$\begin{aligned} \langle w^{\mathrm{f}}_\varepsilon\rangle(\boldsymbol{x}) =& \int_Y w^{\mathrm{f}}_\varepsilon(\boldsymbol{x},\boldsymbol{y})\mathrm{d}\boldsymbol{y} \\ =& \frac{1}{2}\hat{C}_{ijkl}\mathcal{E}_{ij}(\boldsymbol{x})\mathcal{E}_{kl}(\boldsymbol{x}) + \frac{\varepsilon^2}{2}\hat{D}_{ijklmn}\mathcal{E}_{ijk}(\boldsymbol{x})\mathcal{E}_{lmn}(\boldsymbol{x}) \\ &+ \frac{1}{2}\hat{G}(T-T^0)^2(\boldsymbol{x}) - \hat{\beta}_{ij}\mathcal{E}_{ij}(\boldsymbol{x})(T-T^0)(\boldsymbol{x}) + c(T-T^0)(\boldsymbol{x}) - cT\ln\frac{T}{T^0}(\boldsymbol{x}), \end{aligned} \tag{A.10}$$

where the homogenized material parameters are defined in Eq. (15).

Appendix B. Non-negativity of $\hat{\kappa}_{ij}$

In this section, we show that the homogenized thermal conductivity tensor $\hat{\kappa}_{ij}$ defined in Eq. (39) is non-negative definite. Specifically, for any nonzero vector $\boldsymbol{m} \in \mathbb{R}^2$,

$$m_i\hat{\kappa}_{ij}m_j \geq 0. \tag{B.1}$$

To this end, we introduce the auxiliary function associated with the cell function $\boldsymbol{M}$:

$$\tilde{m}_i(\boldsymbol{y}) = m_i + m_k\frac{\partial M_k}{\partial y_i}(\boldsymbol{y}). \tag{B.2}$$

A straightforward calculation yields

$$\begin{aligned} \int_{Y^s}\tilde{m}_i\kappa\tilde{m}_i\,\mathrm{d}\boldsymbol{y} =& \int_{Y^s}\left(m_i + m_p\frac{\partial M_p}{\partial y_i}\right)\kappa\left(m_i + m_j\frac{\partial M_j}{\partial y_i}\right)\mathrm{d}\boldsymbol{y} \\ =& \int_{Y^s} m_i\kappa\left(m_i + m_j\frac{\partial M_j}{\partial y_i}\right)\mathrm{d}\boldsymbol{y} + \int_{Y^s} m_p\frac{\partial M_p}{\partial y_i}\kappa\left(m_i + m_j\frac{\partial M_j}{\partial y_i}\right)\mathrm{d}\boldsymbol{y}. \end{aligned} \tag{B.3}$$

Since the cell functions $\boldsymbol{M}$ satisfy the variational form of the cell problem (see Eq. (8c)),

$$\int_{Y^s}\kappa\left(\delta_{ij} + \frac{\partial M_j}{\partial y_i}\right)\frac{\partial v}{\partial y_i}\mathrm{d}\boldsymbol{y} = 0, \quad \forall j = 1,2,\ \forall v \in \mathcal{D}(Y^s), \tag{B.4}$$

choosing $v = M_p$ gives

$$m_p\int_{Y^s}\frac{\partial M_p}{\partial y_i}\kappa\left(\delta_{ij} + \frac{\partial M_j}{\partial y_i}\right)\mathrm{d}\boldsymbol{y}\,m_j = 0. \tag{B.5}$$

Hence, the second integral in the above expansion vanishes.

By the definition of $\hat{\kappa}_{ij}$ in Eq. (39), the first integral reduces to

$$\int_{Y^s} m_i\kappa\left(m_i + m_j\frac{\partial M_j}{\partial y_i}\right)\mathrm{d}\boldsymbol{y} = m_i\int_{Y^s}\kappa\left(\delta_{ij} + \frac{\partial M_j}{\partial y_i}\right)\mathrm{d}\boldsymbol{y}\,m_j = m_i\hat{\kappa}_{ij}m_j. \tag{B.6}$$

Therefore, combining the above results we arrive at

$$m_i\hat{\kappa}_{ij}m_j = \int_{Y^s}\tilde{m}_i\kappa\tilde{m}_i\,\mathrm{d}\boldsymbol{y} \geq 0, \tag{B.7}$$

since $\kappa > 0$ almost everywhere in Y.

This establishes that the homogenized conductivity tensor $\hat{\kappa}_{ij}$ is non-negative definite.

Data availability

Data will be made available on request.

References

Aifantis, E.C., 1999. Strain gradient interpretation of size effects. Int. J. Fract. 95, 299–314.

Ashby, M.F., Hallam, S., 1986. The failure of brittle solids containing small cracks under compressive stress states. Acta Metall. 34, 497–510.

Bacigalupo, A., De Bellis, M.L., Zavarise, G., 2022. Asymptotic homogenization approach for anisotropic micropolar modeling of periodic cauchy materials. Comput. Methods Appl. Mech. Engrg. 388, 114201.

Bakhvalov, N., Panasenko, G., 1989. Homogenisation: Averaging processes in periodic media. Math. Appl.
Banea, M., Silva, L.D., Campilho, R., 2012. Effect of temperature on tensile strength and mode i fracture toughness of a high temperature epoxy adhesive. J. Adhes. Sci. Technol. 26, 939–953.
Bensoussan, A., Lions, J.L., Papanicolaou, G., Caughey, T., 1979. Asymptotic analysis of periodic structures.
Biswas, S., Jana, P., Chatterjee, A., 2016. Hysteretic damping in an elastic body with frictional microcracks. Int. J. Mech. Sci. 108, 61–71.
Bjerke, T., Lambros, J., 2002. Heating during shearing and opening dominated dynamic fracture of polymers. Exp. Mech. 42, 107–114.
Bjerke, T.W., Lambros, J., 2003. Theoretical development and experimental validation of a thermally dissipative cohesive zone model for dynamic fracture of amorphous polymers. J. Mech. Phys. Solids 51, 1147–1170.
Bjerke, T., Li, Z., Lambros, J., 2002. Role of plasticity in heat generation during high rate deformation and fracture of polycarbonate. Int. J. Plast. 18, 549–567.
Bougaut, O., Rittel, D., 2001. On crack-tip cooling during dynamic crack initiation. Int. J. Solids Struct. 38, 2517–2532.
Challamel, N., 2010. A variationally based nonlocal damage model to predict diffuse microcracking evolution. Int. J. Mech. Sci. 52, 1783–1800.
Chen, J., Wei, Y., Huang, Y.t., Hutchinson, J., Hwang, K., 1999. The crack tip fields in strain gradient plasticity: the asymptotic and numerical analyses. Eng. Fract. Mech. 64, 625–648.
Chen, Y., Xu, Y., Pellegrino, A., 2025. Influence of strain rate and temperature on the multiaxial failure stress locus of a polyamide syntactic foam. Int. J. Mech. Sci. 286, 109864.
Chen, H., Zhao, C., Zhang, R., 2024. Damage mechanism of rock induced by microcrack evolution: A multi-dimensional perspective. Eng. Fract. Mech. 309, 110420.
Cioranescu, D., 1999. An Introduction To Homogenization. Oxford University Press.
Das, A., Kagebein, P., Küchemann, S., Maaß, R., 2018. Temperature rise from fracture in a zr-based metallic glass. Appl. Phys. Lett. 112.
Dascalu, C., Bilbie, G., Agiasofitou, E., 2008. Damage and size effects in elastic solids: A homogenization approach. Int. J. Solids Struct. 45, 409–430.
Dascalu, C., Gbetchi, K., 2019. Dynamic evolution of damage by microcracking with heat dissipation. Int. J. Solids Struct. 174, 128–144.
Dehghani, H., Perrin, H., Belouettar, S., 2024. Effective heat conductivity of composites with stochastic microstructure using asymptotic homogenization. Compos. Struct. 345, 118364.
Delfani, M., Bidi, H., 2024. Strain-gradient solution to elastodynamic scattering from a cylindrical inhomogeneity. Int. J. Mech. Sci. 261, 108667.
Dong, Z., Quan, W., Ma, X., Li, X., Zhou, J., 2023. Asymptotic homogenization of effective thermal-elastic properties of concrete considering its three-dimensional mesostructure. Comput. Struct. 279, 106970.
Erzar, B., Buzaud, E., 2012. Shockless spalling damage of alumina ceramic. Eur. Phys. J. Spec. Top. 206, 71–77.
Fagerström, M., Larsson, R., 2008. A thermo-mechanical cohesive zone formulation for ductile fracture. J. Mech. Phys. Solids 56, 3037–3058.
Fantoni, F., Bacigalupo, A., Paggi, M., Reinoso, J., 2020. A phase field approach for damage propagation in periodic microstructured materials. Int. J. Fract. 223, 53–76.
Feng, G., Kang, Y., Meng, T., Hu, Y.q., Li, X.h., 2017. The influence of temperature on mode i fracture toughness and fracture characteristics of sandstone. Rock Mech. Rock Eng. 50, 2007–2019.
Feng, G., Wang, X.c., Kang, Y., Luo, S.g., Hu, Y.q., 2019. Effects of temperature on the relationship between mode-i fracture toughness and tensile strength of rock. Appl. Sci. 9, 1326.
François, B., Dascalu, C., 2010. A two-scale time-dependent damage model based on non-planar growth of micro-cracks. J. Mech. Phys. Solids 58, 1928–1946.
Freund, L.B., 1990. Dynamic Fracture Mechanics. Cambridge University Press.
Fuller, K., Fox, P., Field, J.E., 1975. The temperature rise at the tip of fast-moving cracks in glassy polymers. Proc. R. Soc. A 341, 537–557.
Funatsu, T., Seto, M., Shimada, H., Matsui, K., Kuruppu, M., 2004. Combined effects of increasing temperature and confining pressure on the fracture toughness of clay bearing rocks. Int. J. Rock Mech. Min. Sci. 41, 927–938.
Gambarotta, L., Lagomarsino, S., 1993. A microcrack damage model for brittle materials. Int. J. Solids Struct. 30, 177–198.
Grah, M., Alzebdeh, K., Sheng, P., Vaudin, M., Bowman, K., Ostoja-Starzewski, M., 1996. Brittle intergranular failure in 2d microstructures: experiments and computer simulations. Acta Mater. 44, 4003–4018.
Grassl, P., Grégoire, D., Solano, L.R., Pijaudier-Cabot, G., 2012. Meso-scale modelling of the size effect on the fracture process zone of concrete. Int. J. Solids Struct. 49, 1818–1827.
Griffith, A.A., 1921. Vi. the phenomena of rupture and flow in solids. Philos. Trans. R. Soc. Lond. Ser. A, Contain. Pap. A Math. Or Phys. Character 221, 163–198.
Gross, D., Seelig, T., 2006. Fracture mechanics: with an introduction to micromechanics. Springer.
Irwin, G.R., 1958. Fracture. Elast. Plasticity/Elastizität Und Plastizität 551–590.
Ivančević, D., Ratković, J., 2025. Strain rate effects on composite impact damage: Bvid to perforation. Int. J. Mech. Sci. 110907.
Keita, O., Dascalu, C., François, B., 2014. A two-scale model for dynamic damage evolution. J. Mech. Phys. Solids 64, 170–183.
Kipp, M., Grady, D., Chen, E., 1980. Strain-rate dependent fracture initiation. Int. J. Fract. 16, 471–478.
Knauss, W., Ravi-Chandar, K., 1985. Some basic problems in stress wave dominated fracture. Int. J. Fract. 27, 127–143.
Kranz, R.L., 1983. Microcracks in rocks: a review. Tectonophysics 100, 449–480.
Kuhn, C., Müller, R., 2009. Phase field simulation of thermomechanical fracture. In: PAMM: Proceedings in Applied Mathematics and Mechanics. Wiley Online Library, pp. 191–192.
Lam, D.C., Yang, F., Chong, A., Wang, J., Tong, P., 2003. Experiments and theory in strain gradient elasticity. J. Mech. Phys. Solids 51, 1477–1508.
Lambert, D.E., Ross, C.A., 2000. Strain rate effects on dynamic fracture and strength. Int. J. Impact Eng. 24, 985–998.
Lazar, M., 2024. Dislocations in nonlocal simplified strain gradient elasticity: Eringen meets aifantis. Int. J. Mech. Sci. 275, 109294.
Li, J., 2011. A micromechanics-based strain gradient damage model for fracture prediction of brittle materials–part i: Homogenization methodology and constitutive relations. Int. J. Solids Struct. 48, 3336–3345.
Li, X., Li, H., Zhang, Q., Jiang, J., Zhao, J., 2018. Dynamic fragmentation of rock material: characteristic size, fragment distribution and pulverization law. Eng. Fract. Mech. 199, 739–759.
Li, J., Pham, T., Abdelmoula, R., Song, F., Jiang, C., 2011. A micromechanics-based strain gradient damage model for fracture prediction of brittle materials–part ii: Damage modeling and numerical simulations. Int. J. Solids Struct. 48, 3346–3358.
Li, Q., Rao, Y., Tian, X., Chen, J., Yu, X., Cui, J., Xiang, M., 2025a. An energetically consistent two-scale asymptotic approach for modeling dynamic deformation and fracture of brittle materials. Eur. J. Mech. A Solids 105934.
Li, Q., Rao, Y., Yang, Z., Cui, J., Xiang, M., 2025b. Asymptotic homogenization-based strain gradient elastodynamics: Governing equations, well-posedness and numerical examples. Comput. Methods Appl. Mech. Engrg. 442, 118010.
Liu, S., Hao, S., Shen, Y., 2024. Asymptotic homogenization for phase field fracture of heterogeneous materials and application to toughening. Compos. Struct. 339, 118134.
Ma, P.S., Liu, X.C., Luo, X.L., Li, S., Zhang, L.W., 2024. Asymptotic homogenization of phase-field fracture model: An efficient multiscale finite element framework for anisotropic fracture. Internat. J. Numer. Methods Engrg. 125, e7489.
Mason, J., Rosakis, A., Ravichandran, G., 1994. On the strain and strain rate dependence of the fraction of plastic work converted to heat: an experimental study using high speed infrared detectors and the kolsky bar. Mech. Mater. 17, 135–145.
Meguid, S., Gong, S., Gaultier, P., 1991. Main crack-microcrack interaction under mode i, ii and iii loadings: Shielding and amplification. Int. J. Mech. Sci. 33, 351–359.
Miehe, C., Hofacker, M., Schänzel, L.M., Aldakheel, F., 2015a. Phase field modeling of fracture in multi-physics problems. part ii. coupled brittle-to-ductile failure criteria and crack propagation in thermo-elastic–plastic solids. Comput. Methods Appl. Mech. Engrg. 294, 486–522.
Miehe, C., Schaenzel, L.M., Ulmer, H., 2015b. Phase field modeling of fracture in multi-physics problems. part i. balance of crack surface and failure criteria for brittle crack propagation in thermo-elastic solids. Comput. Methods Appl. Mech. Engrg. 294, 449–485.
Moore, D.E., Lockner, D., 1995. The role of microcracking in shear-fracture propagation in granite. J. Struct. Geol. 17, 95–114.
Nemat-Nasser, S., Obata, M., 1988. A microcrack model of dilatancy in brittle materials. J. Appl. Mech. 55, 24.
Nirmalendran, S., Horii, H., 1992. Analytical modeling of microcracking and bridging in the fracture of quasi-brittle materials. J. Mech. Phys. Solids 40, 863–886.
Oleïnik, O.A., Shamaev, A., Yosifian, G., 1992. Mathematical problems in elasticity and homogenization, vol. 26, Elsevier.
Ožbolt, J., Sharma, A., İrhan, B., Sola, E., 2014. Tensile behavior of concrete under high loading rates. Int. J. Impact Eng. 69, 55–68.
Paliwal, B., Ramesh, K., 2008. An interacting micro-crack damage model for failure of brittle materials under compression. J. Mech. Phys. Solids 56, 896–923.
Pandey, K.N., Chand, S., 2008. Analysis of temperature distribution near the crack tip under constant amplitude loading. Fatigue Fract. Eng. Mater. Struct. 31, 316–326.
Pandya, K.S., Roth, C.C., Mohr, D., 2020. Strain rate and temperature dependent fracture of aluminum alloy 7075: Experiments and neural network modeling. Int. J. Plast. 135, 102788.
Peerlings, R.H., de Borst, R., Brekelmans, W.M., de Vree, J., 1996. Gradient enhanced damage for quasi-brittle materials. Internat. J. Numer. Methods Engrg. 39, 3391–3403.
Pitchai, P., Harursampath, D., Guruprasad, P., 2023. Stochastic analysis of a three phase composite using variational asymptotic based homogenization technique coupled with stochastic reduced order model. Compos. Struct. 316, 117040.
Placidi, L., Barchiesi, E., 2018. Energy approach to brittle fracture in strain-gradient modelling. Proc. R. Soc. A: Math. Phys. Eng. Sci. 474, 20170878.
Qingliang, S., Guangmeng, Y., Caixiang, X., Qiang, S., Hejun, L., Xiyuan, Y., Fu, M., 2023. A microcracking-based model for the dynamic failure of carbon/carbon composites. Int. J. Mech. Sci. 260, 108625.
Ramalingam, K., Motaman, S.A.H., Haase, C., Krupp, U., 2025. Thermo-micro-mechanical modeling of plasticity and damage in single-phase s700 steel. Int. J. Mech. Sci. 286, 109909.

Rao, Y., Xiang, M., Cui, J., 2022. A strain gradient brittle fracture model based on two-scale asymptotic analysis. J. Mech. Phys. Solids 159, 104752.
Rao, Y., Xiang, M., Li, Q., Cui, J., 2023. A unified two-scale theory for modeling microstructural length scale, strain gradient and strain rate effects on brittle fracture. Int. J. Solids Struct. 268, 112176.
Ravi-Chandar, K., Knauss, W., 1984a. An experimental investigation into dynamic fracture: Ii. microstructural aspects. Int. J. Fract. 26, 65–80.
Ravi-Chandar, K., Knauss, W.G., 1984b. An experimental investigation into dynamic fracture: I. crack initiation and arrest. Int. J. Fract. 25, 247–262.
Ravi-Chandar, K., Knauss, W.G., 1984c. An experimental investigation into dynamic fracture: Iii. on steady-state crack propagation and crack branching. Int. J. Fract. 26, 141–154.
Ravi-Chandar, K., Yang, B., 1997. On the role of microcracks in the dynamic fracture of brittle materials. J. Mech. Phys. Solids 45, 535–563.
Reinhardt, H., Weerheijm, J., 1991. Tensile fracture of concrete at high loading rates taking account of inertia and crack velocity effects. Int. J. Fract. 51, 31–42.
Rice, J.R., 1968. A path independent integral and the approximate analysis of strain concentration by notches and cracks. Technical Report.
Rittel, D., 2000. Experimental investigation of transient thermoplastic effects in dynamic fracture. Int. J. Solids Struct. 37, 2901–2913.
Ruan, H., Rezaei, S., Yang, Y., Gross, D., Xu, B.X., 2023. A thermo-mechanical phase-field fracture model: Application to hot cracking simulations in additive manufacturing. J. Mech. Phys. Solids 172, 105169.
Shang, M., Yang, H., 2025. Strain rate dependent ductile fracture behavior of q960 ultra-high-strength steel. Int. J. Mech. Sci. 293, 110183.
Shen, Y., Ma, T., Li, J., 2025. Investigation of dynamic fractures under varying stress states. Int. J. Mech. Sci. 293, 110177.
Smyshlyaev, V.P., Cherednichenko, K.D., 2000. On rigorous derivation of strain gradient effects in the overall behaviour of periodic heterogeneous media. J. Mech. Phys. Solids 48, 1325–1357.
Srivastava, C., Mahesh, V., Pitchai, P., Guruprasad, P., Petrinic, N., Scarpa, F., Harursampath, D., Ponnusami, S.A., 2023. Effective mechanical properties of auxetic materials: Numerical predictions using variational asymptotic method based homogenization. J. Appl. Mech. 90, 111001.
Stolkarts, V., Keer, L., Fine, M., 1999. Damage evolution governed by microcrack nucleation with application to the fatigue of 63sn–37pb solder. J. Mech. Phys. Solids 47, 2451–2468.
Sun, N., Hsu, T., 1996. Thermomechanical coupling effects on fractured solids. Int. J. Fract. 78, 67–87.
Tangella, R.G., Kumbhar, P., Annabattula, R.K., 2022. Hybrid phase-field modeling of thermo-elastic crack propagation. Int. J. Comput. Methods Eng. Sci. Mech. 23, 29–44.
Verleysen, P., Peirs, J., 2017. Quasi-static and high strain rate fracture behaviour of ti6al4v. Int. J. Impact Eng. 108, 370–388.
Wang, H., Ramesh, K., 2004. Dynamic strength and fragmentation of hot-pressed silicon carbide under uniaxial compression. Acta Mater. 52, 355–367.
Weichert, R., Schönert, K., 1974. On the temperature rise at the tip of a fast running crack. J. Mech. Phys. Solids 22, 127–133.
Weichert, R., Schönert, K., 1978. Heat generation at the tip of a moving crack. J. Mech. Phys. Solids 26, 151–161.
Wrzesniak, A., Dascalu, C., Bésuelle, P., 2015. A two-scale time-dependent model of damage: Influence of micro-cracks friction. Eur. J. Mech. A Solids 49, 345–361.
Xiang, M., Jiang, S., Cui, J., Xu, Y., Chen, J., 2021. Coupling of dynamic ductile damage and melting in shock-induced micro-spalling: Modeling and applications. Int. J. Plast. 136, 102849.
Xiang, Y., Song, M., Zhao, H., Zhu, Q., Jin, M., Liu, J., 2025. A novel two-scale microcrack band damage model for quasi-brittle materials. J. Mech. Phys. Solids 106304.
Yang, Z., Cui, J., Wu, Y., Wang, Z., Wan, J., 2015. Second-order two-scale analysis method for dynamic thermo-mechanical problems in periodic structure. Int. J. Numer. Anal. Model. 12, 144–161.
Yang, B., Zine, A., Droz, C., Ichchou, M., 2022. Two-dimensional periodic structures modeling based on second strain gradient elasticity for a beam grid. Int. J. Mech. Sci. 222, 107199.
You, H., Xiang, M., Jing, Y., Guo, L., Yang, Z., 2025. A strain gradient phase field model for heterogeneous materials based on two-scale asymptotic homogenization. J. Mech. Phys. Solids 200, 106104.
Zhang, B., Li, H., Kong, L., Shen, H., Zhang, X., 2020. Coupling effects of surface energy, strain gradient, and inertia gradient on the vibration behavior of small-scale beams. Int. J. Mech. Sci. 184, 105834.
Zhong, Y., Qin, W., Yu, W., Zhou, X., Jiao, L., 2015. Variational asymptotic homogenization of magneto-electro-elastic materials with coated fibers. Compos. Struct. 133, 300–311.
Zhou, F., Molinari, J.F., Ramesh, K., 2006. Effects of material properties on the fragmentation of brittle materials. Int. J. Fract. 139, 169–196.
Zinszner, J., Erzar, B., Forquin, P., Buzaud, E., 2015. Dynamic fragmentation of an alumina ceramic subjected to shockless spalling: an experimental and numerical study. J. Mech. Phys. Solids 85, 112–127.